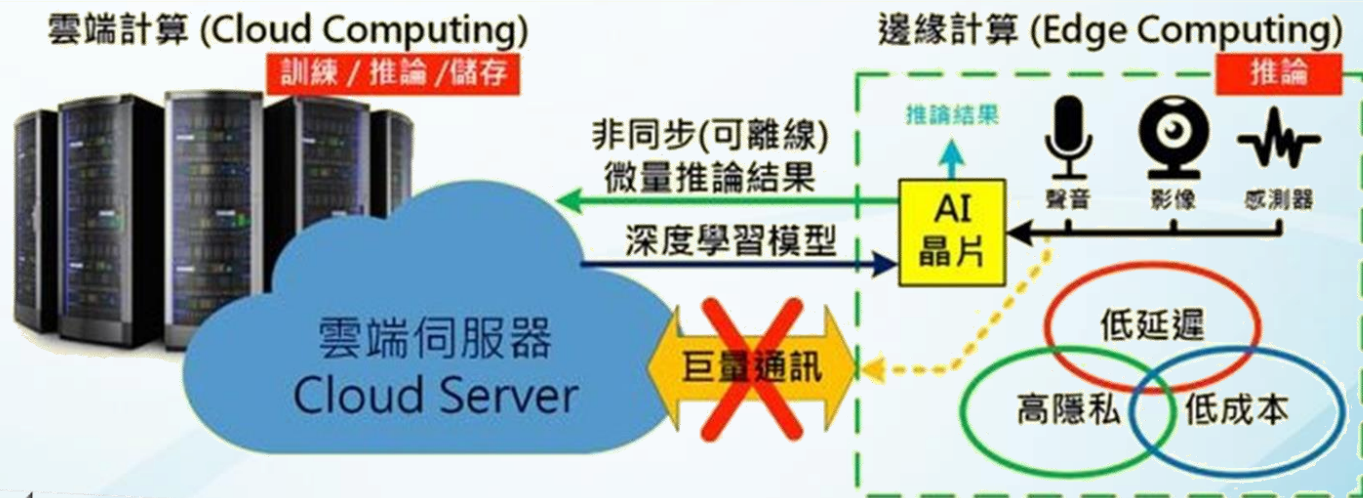
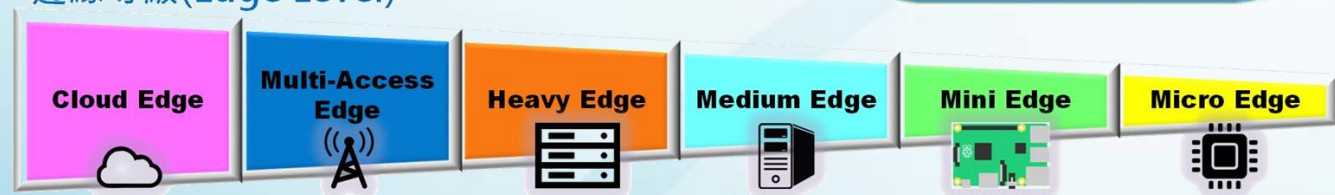


邊緣人工智慧實務 (EE5354701)



Edge AI

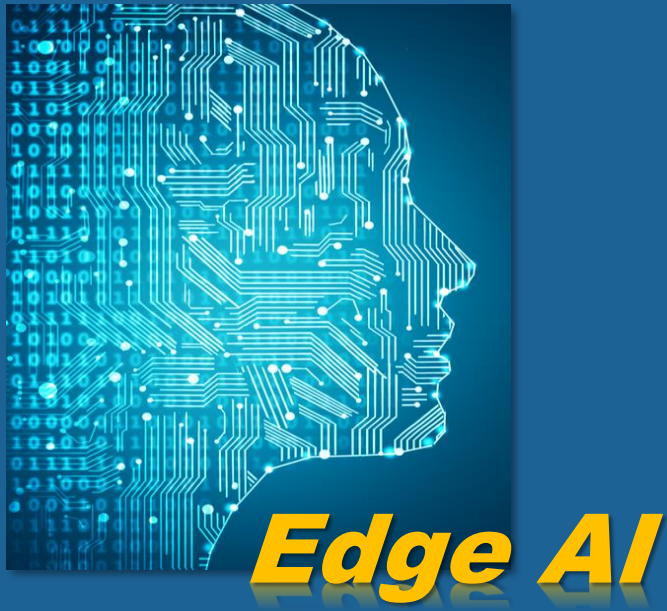
邊緣等級(Edge Level)



4. 開源模型推論工具

電機工程系 人工智慧應用產碩專班 許哲豪 兼任助理教授

簡報大綱

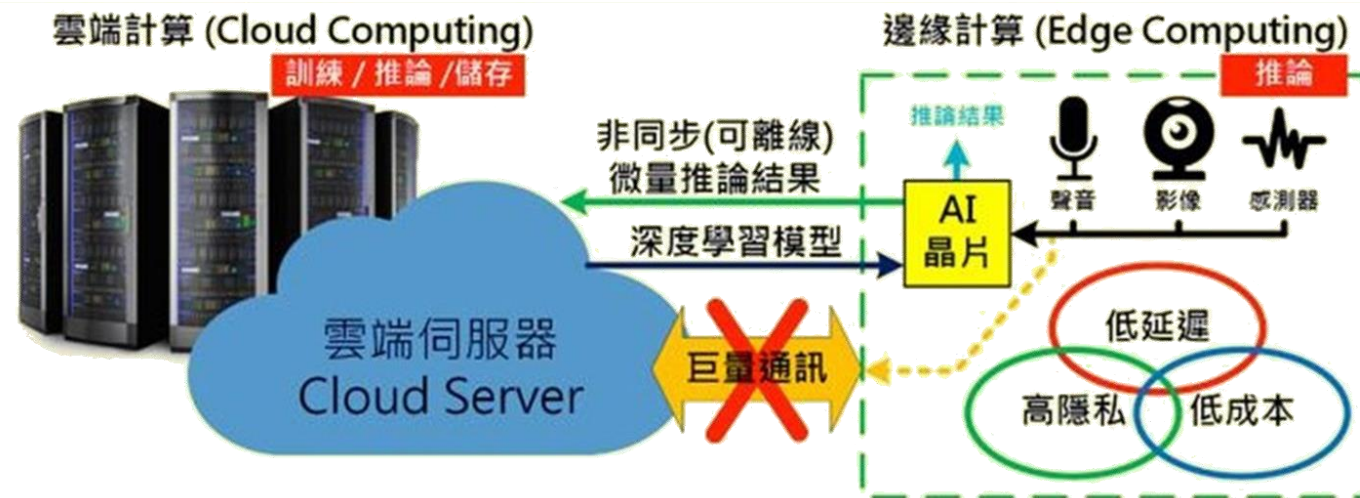


- **4.1. Arm CMSIS**
- **4.2. Edge Impulse**
- **4.3. Intel OpenVINO**

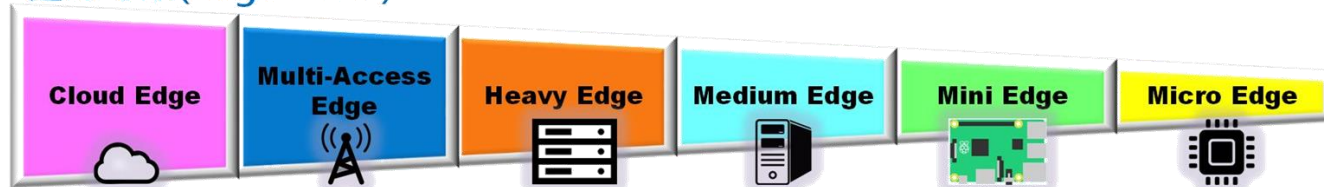
https://hackmd.io/@OmniXRI-Jack/NTUST_EdgeAI_2026



Edge AI

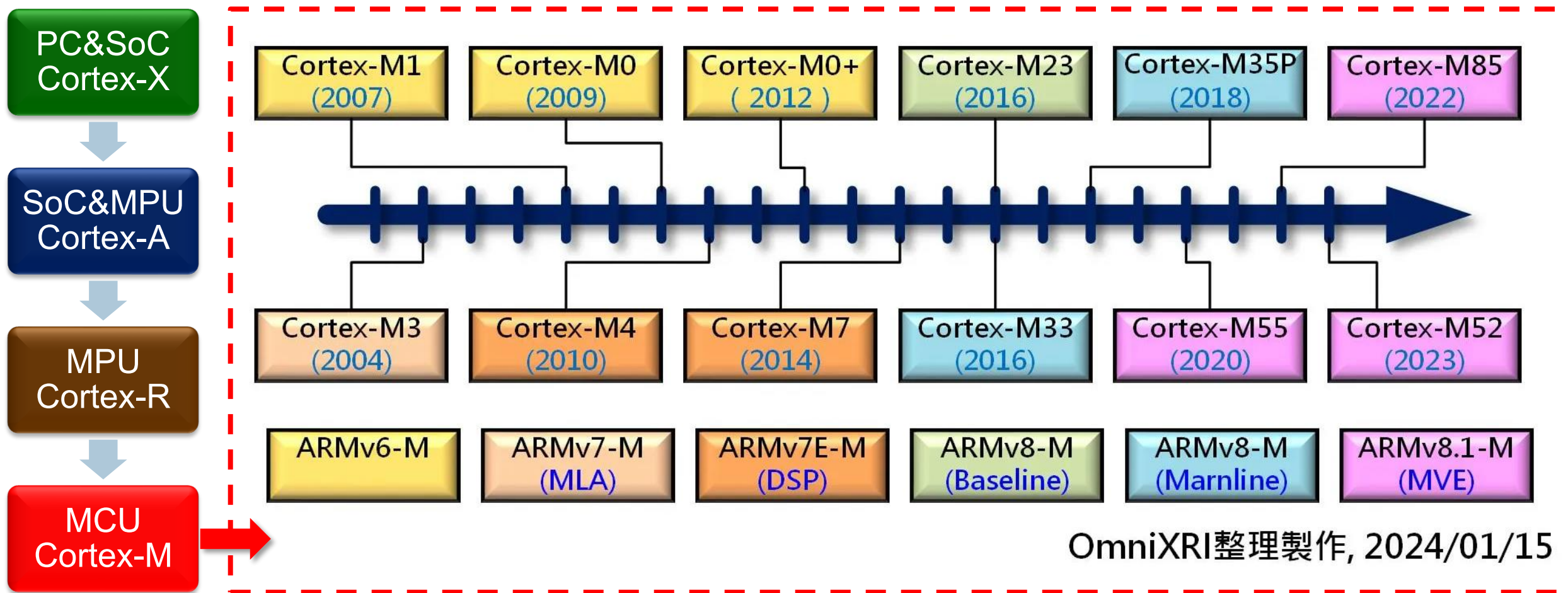


邊緣等級(Edge Level)



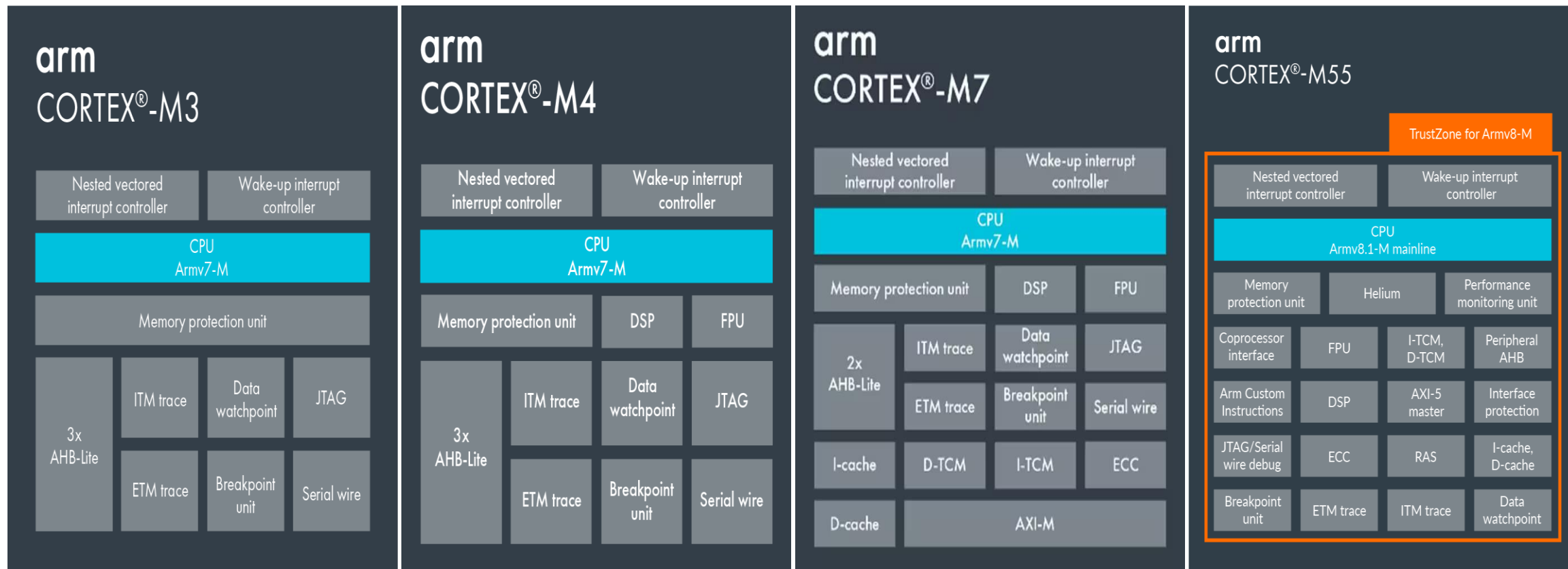
4.1. ARM CMSIS

常見 Arm 晶片CPU等級及指令集



資料來源：<https://omnixri.blogspot.com/2024/01/vmaker-edge-ai-13-npuai.html>

arm Cortex-M 架構差異

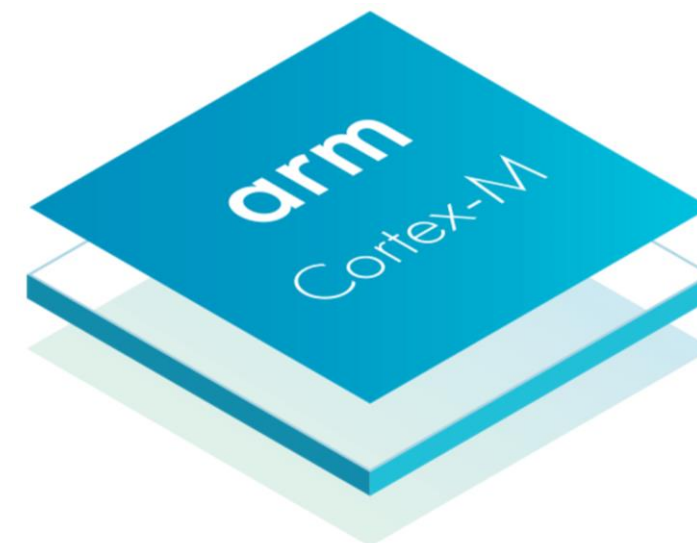


**MCU 的多樣性
造成開發的問題**

矽 智 財：內核不同、週邊不同、記憶體位址不同、安全性不同 ...
實體晶片：封裝不同、接腳數不同、電氣性能不同 ...
開發工具：IDE不同、程式語言不同、編譯器不同 ...

Cortex-M 指令加速操作比較

處理器	指令集	OPs at 100MHz
Cortex-M3	多功能指令 (MLA, 2個OPS) 需要2個週期。 由於處理器還需要執行NN處理的記憶體負載操作，假設Mac vs Load的比率為1:1，因此平均操作/週期為0.6。	0.06 GOPs/sec
Cortex-M4, Cortex-M33	DSP/SIMD指令支持雙MAC操作 (4個操作)。 由於處理器還需要執行NN處理的記憶體加載操作，假設Mac vs Load的比率為1:1，因此平均操作/週期為2。	0.2 GOPs/sec
Cortex-M7	該處理器支持DSP/SIMD和記憶體負載的雙重問題， 因此平均操作/週期為4。	0.4 GOPs/sec
Cortex-M52	使用Helium技術，這些處理器可以與數據負載並聯處理4個MAC/週期。結果，平均操作/週期為8。	0.8 GOPs/sec
Cortex-M55, Cortex-M85	使用Helium技術，這些處理器可以與數據負載並聯處理8個MAC/循環。結果，平均操作/週期為16。	1.6 GOPs/sec

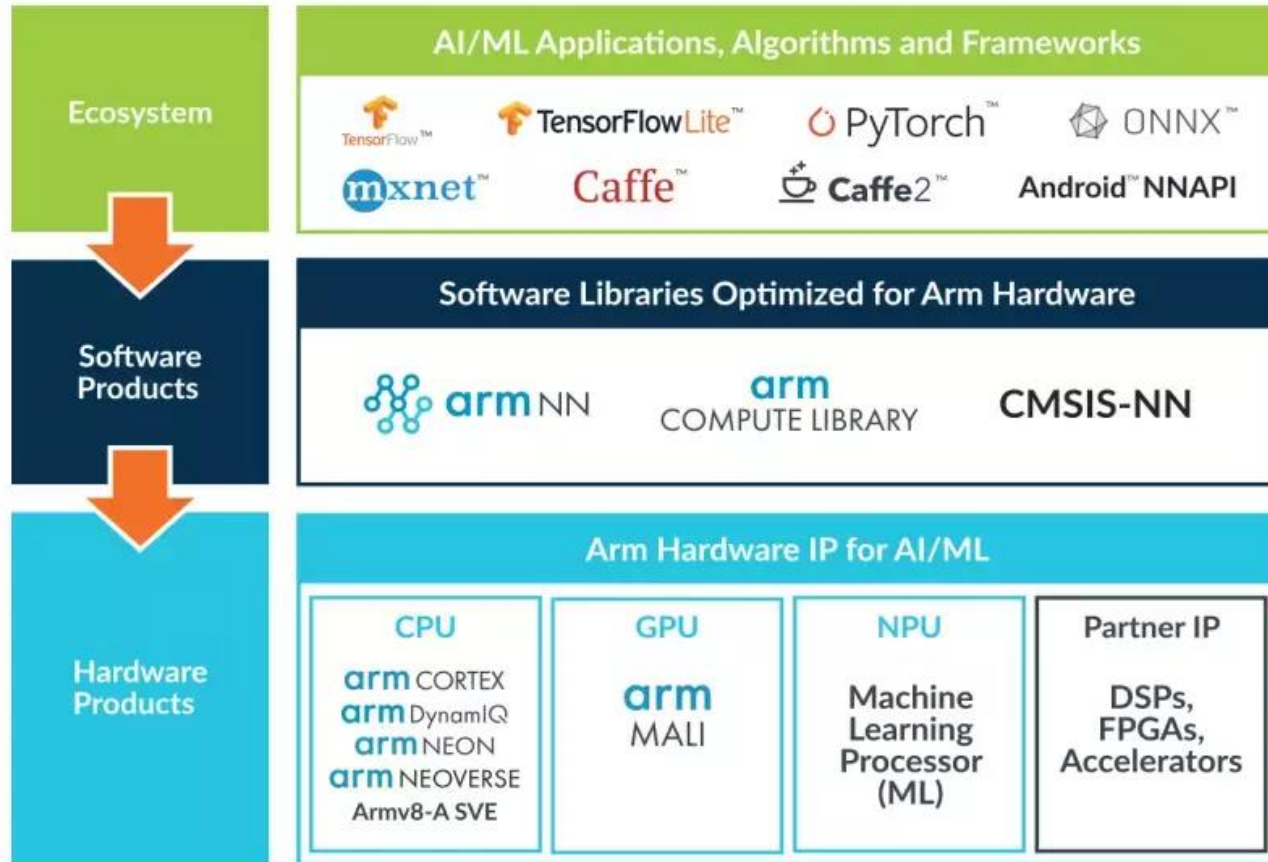


arm v8.1m (Helium) 指令集
M-Profile Vector Extension, MVE

資料來源：https://hackmd.io/@OmniXRI-Jack/arm_developer_cortexm55_ethosu55_guide

Edge AI & TinyML 通用 / 專用開發工具

Arm Based MCU / MPU / GPU / NPU 通用型



Cortex-A Kleidi AI

Vela Compiler

MCU 專用 (半通用) 型

- ST : CubeAI, Nano Edge, Deeplite
- NXP : eIQ, Kinara
- Renesas : Reality AI
- Nordic : Atlazo, Neuton
- TDK : Qeexo
- Nuvoton : NuEdgeWise
- QuickLogic : SensiML
- Infineon : Imagimob
- Qualcomm: Edge Impulse, Arduino APP Lib

何謂 arm CMSIS

arm Cortex 微控制器 (單晶片) 軟體介面標準

(Cortex Microcontroller Software Interface Standard, CMSIS)

- 供應商無關之硬體抽象層，提供統一的外部設備 (Peripheral, 簡稱外設) 軟體介面。
- 符合 ANSI C (C99) 及 C++ (C++03) 語法，支援多種編譯器。
- 提升軟體重用性及移植性、加快學習時間、降低開發成本。



CMSIS-5



CMSIS

Version 5.9.0

Common Microcontroller Software Interface Standard

General

Core(A)

Core(M)

Driver

DSP

NN

RTOS v1

RTOS v2

Pack

Build

SVD

DAP

Zone

Main Page

Usage and Description

▼ CMSIS

▼ Introduction

CMSIS Components

Motivation

Coding Rules

Validation

License

CMSIS Software Pack

Revision History

Introduction

The **CMSIS** is a set of tools, APIs, frameworks, and work flows that help to simplify software re-use, reduce the learning curve for microcontroller developers, speed-up project build and debug, and thus reduce the time to market for new applications.

CMSIS started as a vendor-independent hardware abstraction layer Arm® Cortex®-M based processors and was later extended to support entry-level Arm Cortex-A based processors. To simplify access, CMSIS defines generic tool interfaces and enables consistent device support by providing simple software interfaces to the processor and the peripherals.

CMSIS is defined in close cooperation with various silicon and software vendors and provides a common approach to interface to peripherals, real-time operating systems, and middleware components. It is intended to enable the combination of software components from multiple vendors.

CMSIS is open-source and collaboratively developed on [GitHub](#).

CMSIS Components

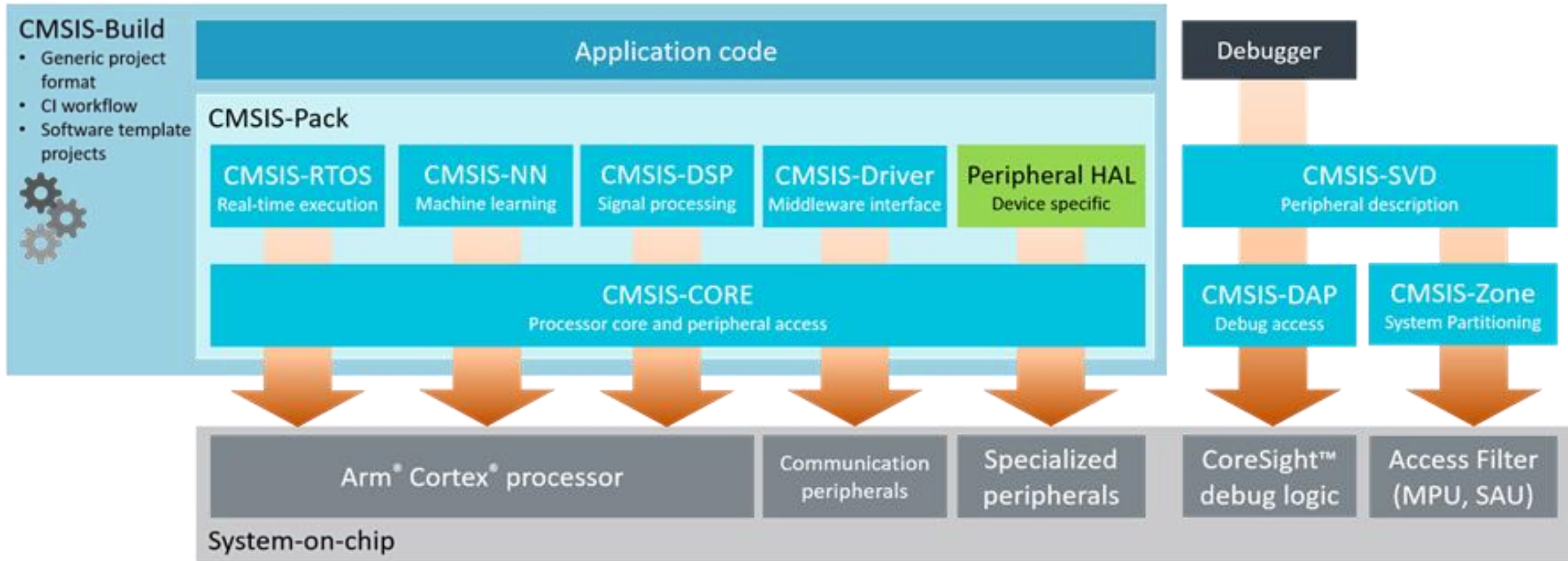
CMSIS-...	Target Processors	Description
Core(M)	All Cortex-M, SecurCore	Standardized API for the Cortex-M processor core and peripherals. Includes intrinsic functions for Cortex-M4/M7/M33/M35P SIMD instructions.
Core(A)	Cortex-A5/A7/A9	Standardized API and basic run-time system for the Cortex-A5/A7/A9 processor core and peripherals.
Driver	All Cortex	Generic peripheral driver interfaces for middleware. Connects microcontroller peripherals with middleware that implements for example communication stacks, file systems, or graphic user interfaces.
DSP	All Cortex-M	DSP library collection with over 60 functions for various data types: fixed-point (fractional q7, q15, q31) and single precision floating-point (32-bit). Implementations optimized for the SIMD instruction set are available for Cortex-M4/M7/M33/M35P.

(2022/5/2 發行)

Core(A)
Core(M)
Driver
DSP
NN
RTOS v1
RTOS v2
Pack
Build
SVD
DAP
Zone

https://arm-software.github.io/CMSIS_5/latest/General/html/index.html

CMSIS-5 架構圖



資料來源：https://arm-software.github.io/CMSIS_5/latest/General/html/index.html

CMSIS-6

CMSIS Version 6.1.0
Common Microcontroller Software Interface Standard

V6.0.0 2023/12/8 發行

Overview Core Driver RTOS2 DSP NN View Compiler Toolbox Stream DAP Zone

▼ CMSIS

▼ Introduction

- CMSIS Components
- Benefits
- Development
- Coding Rules
- Validation
- Migration from CMSIS v5
- License
- CMSIS Software Pack
- Revision History

CMSIS Base Software Components

- Provide software abstractions for basic level functionalities of a device.
- Maintained in the same GitHub repository and delivered as one **CMSIS Software Pack** with the name `Arm::CMSIS`.

CMSIS-Core

Standardized access to Arm Cortex processor cores

[Guide](#) | [GitHub](#) | [Pack](#)

CMSIS-Driver

Generic peripheral driver interfaces for middleware

[Guide](#) | [GitHub](#) | [Pack](#)

CMSIS-RTOS2

Common API for real-time operating systems

[Guide](#) | [GitHub](#) | [Pack](#)

CMSIS Extended Software Components

- Implement specific functionalities optimized for execution on Arm processors.
- Maintained in separate GitHub repositories and delivered in standalone CMSIS-Packs.

CMSIS-DSP

Optimized compute functions

CMSIS-NN

Efficient and performant neural

CMSIS-View

Event Recorder and Component

Core
Driver
RTOS2
DSP
NN
View
Compiler
Toolbox
Stream
DAP
Zone

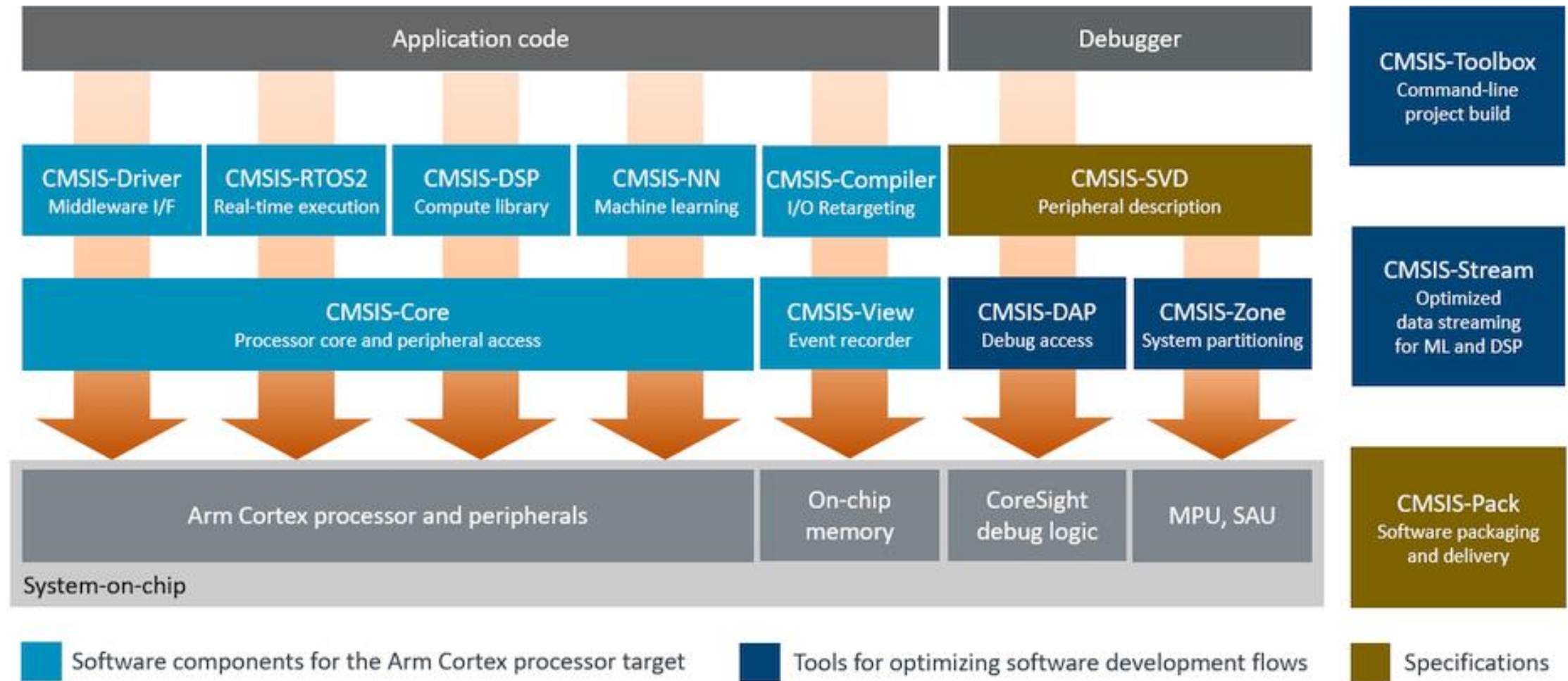
https://arm-software.github.io/CMSIS_6/latest/General/index.html

CMSIS-6 主要元件功能

元件名稱	功能說明	元件名稱	功能說明
Core	對Cortex內核的標準化處理	Toolbox	一套用於套件的命令行工具
Driver	中間層通用外設驅動介面	Stream	用於優化DSP/ML區塊數據流工具
RTOS2	即時作業系統通用API	DAP	連接到除錯埠除錯之韌體
DSP	嵌入式系統最佳化計算函式	Zone	定義用於描述系統並分區的方法
NN	高效及高性能神經網路內核	Pack	軟體元件和設備支援交付機制
View	事件記錄器及元件觀察器技術	SVD	用於除錯設備的外圍設備描述
Compiler	重定向標準C運行時庫的I/O函數		

資料來源：https://arm-software.github.io/CMSIS_6/latest/General/index.html

CMSIS-6 架構圖



資料來源：https://arm-software.github.io/CMSIS_6/latest/General/index.html

CMSIS 5 / 6 Github 比較

CMSIS_5 / CMSIS /
..
Core
CoreValidation
Core_A
DAP/Firmware
DSP
Documentation

DoxyGen
Driver
NN
Pack
RTOS
RTOS2
Utilities

CMSIS_6 / CMSIS /

..
Core
CoreValidation
Documentation
Driver
RTOS2

https://github.com/ARM-software/CMSIS_5/tree/develop/CMSIS

https://github.com/ARM-software/CMSIS_6/tree/main/CMSIS

CMSIS 5 / 6 主要差異

項目	CMSIS-5	CMSIS-6
支持硬體	Cortex-M0,M0+,M3,M4,M7	新增Cortex-M23,M33,M55,M85
安全特性	不支持 TrustZone	原生支持 TrustZone
CMSIS-Pack改進	基礎功能	改進和MDK及其它IDE的整合
DSP & ML	提供基礎DSP, NN功能	擴展NN庫，滿足更複雜模型
工具鏈和兼容性	支援 GCC, IAR, Keil	改善現代工具鏈兼容性便於遷移
配置文件和標準化	基礎設備頭文件及啟動碼	改善設備配置文件更靈活應用

註：大多數應用使用 CMSIS-5 就能滿足，若要應用於 MCU AI 則建議改用 CMSIS-6

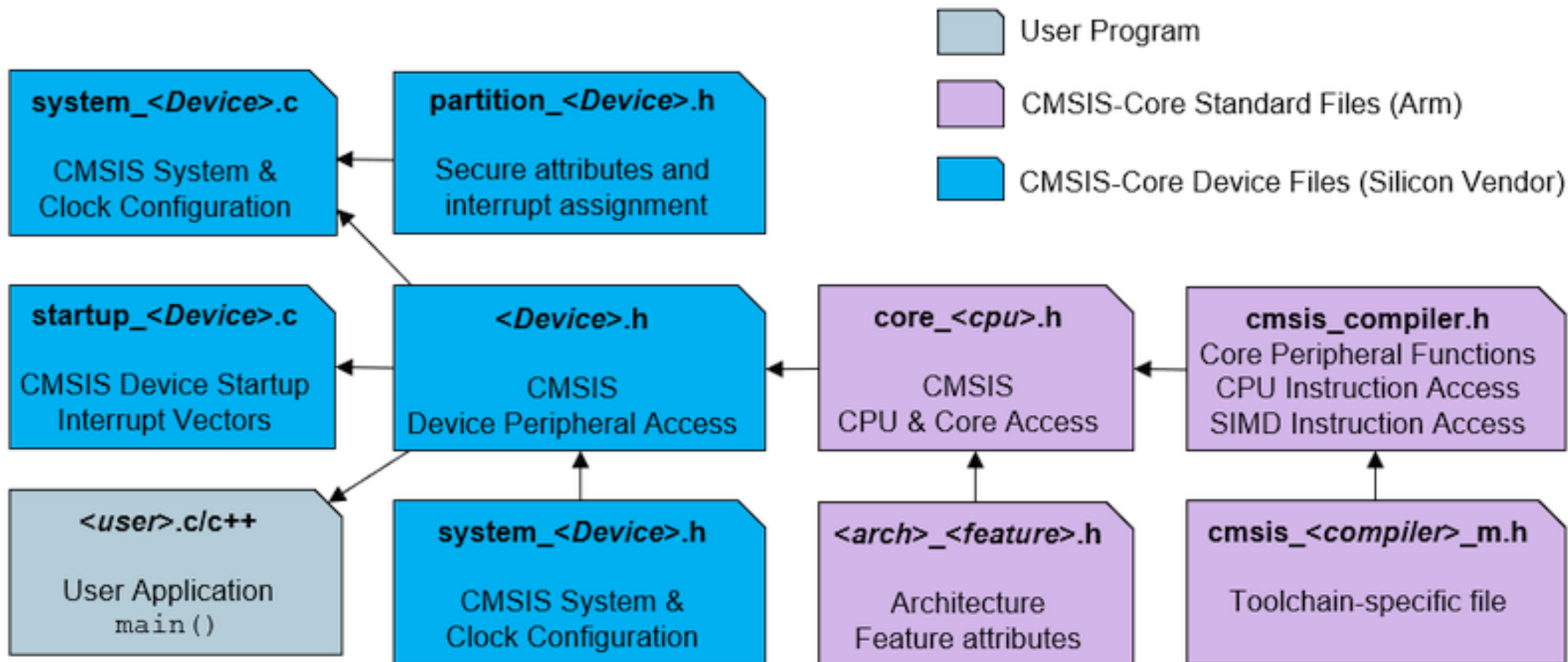
CMSIS-6 Core

CMSIS-Core (Cortex-M) 元件實現了 Arm Cortex-M 設備的基本運行時系統，並允許使用者訪問處理器內核(Core)和週邊裝置(Peripherals)。

Directory	Content
 CMSIS	CMSIS Base software components folder
 Documentation/html/Core	A local copy of this CMSIS-Core (M) documentation
L  Core	CMSIS-Core files
 Include	CMSIS-Core Processor Files.
L  m-profile	Header files specific for Arm M-Profile. See CMSIS-Core Compiler Files and CMSIS-Core Architecture Feature Files.
L  Template	Device Template Files

資料來源：https://arm-software.github.io/CMSIS_6/latest/Core/index.html

CMSIS-6 Core 相關檔案



資料來源：https://arm-software.github.io/CMSIS_6/latest/Core/index.html

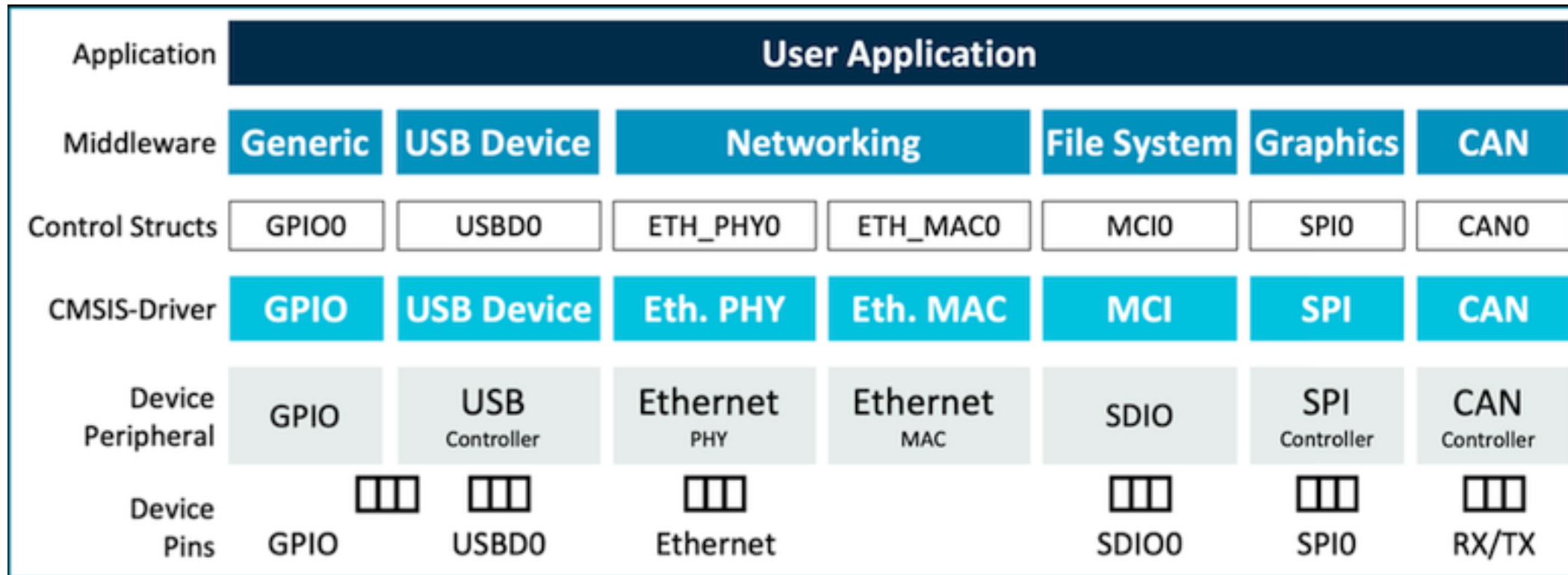
CMSIS-6 Driver

使用者應用程式和中間件元件可以使用 CMSIS-Driver API 控制此類週邊設備，從而在各種生態系統中實現更好的代碼重用和更簡單的集成。

Directory	Content
 CMSIS	CMSIS Base software components folder
 Documentation/html/Driver	A local copy of this CMSIS-Driver documentation
 Driver	Directory with CMSIS-Driver component, see CMSIS-Driver Files
 DriverTemplates	Driver Template files (<i>Driver_interface.c</i>)
 Include	API header files (<i>Driver_interface.h</i> , <i>Driver_Common.h</i>)
 VIO	Implementation of virtual Input/Output interface (VIO)

資料來源：https://arm-software.github.io/CMSIS_6/latest/Driver/index.html

CMSIS-6 Driver 架構圖

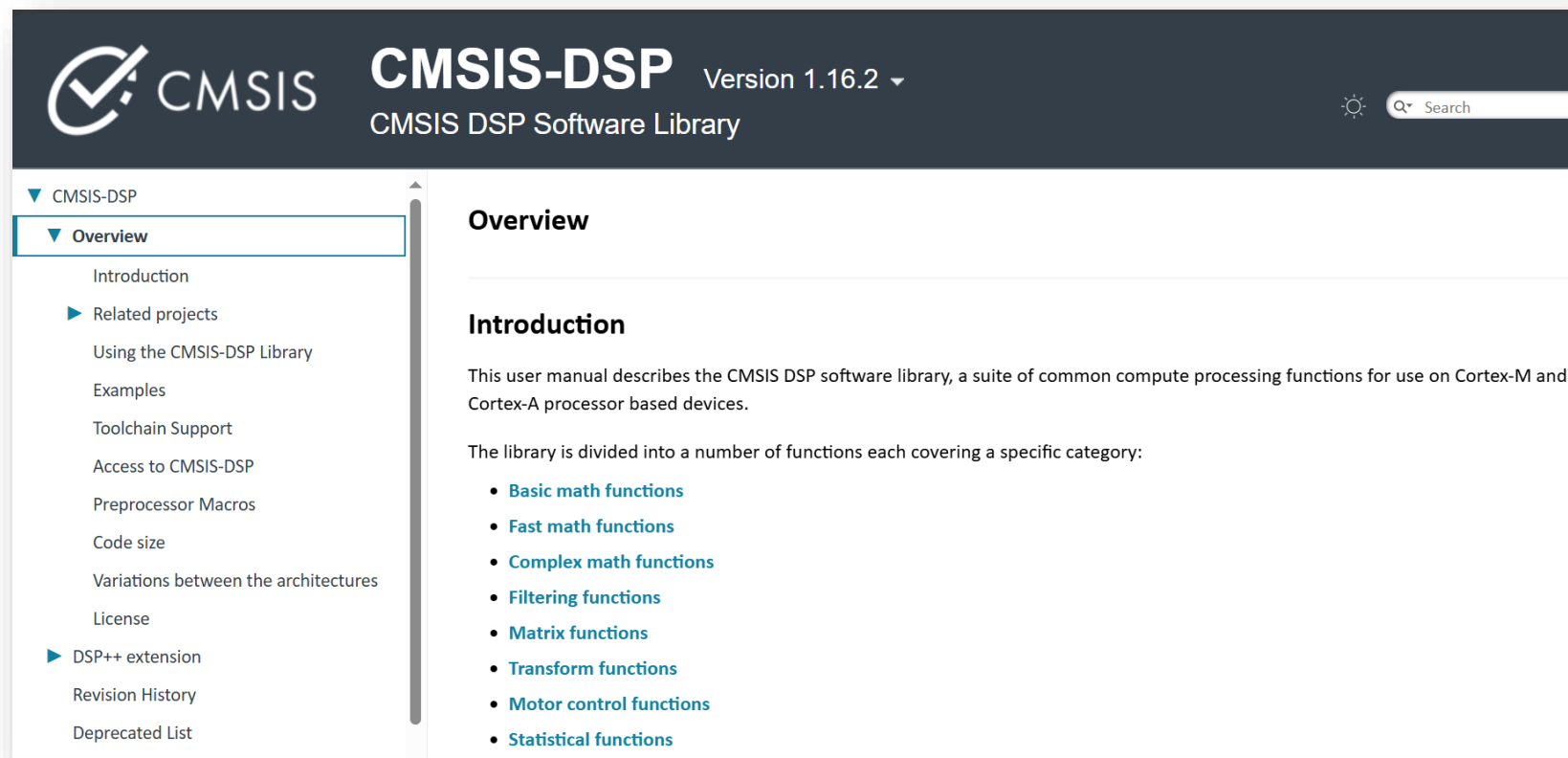


包括 CAN, Ethernet, I2C, MCI, NAND, Flash, SAI, API, Storage, USART, USB, GPIO, VIO, WiFi 等驅動定義

資料來源：https://arm-software.github.io/CMSIS_6/latest/Driver/index.html

CMSIS-6 DSP

CMSIS-DSP 是一個開源軟體庫（在Github已獨立），可實現針對 Arm Cortex-M 和 Cortex-A 處理器進行了優化的常見計算處理功能。



- 基本數學函數
- 快速數學函數
- 複數數學函數
- 濾波器函數
- 矩陣函數
- 變換函數
- 馬達控制函數
- 統計函數
- 支援函數
- 插值函數
- 支持向量機函數
- 貝葉斯分類器函數
- 距離函數
- 四元數函數
- 視窗函數

資料來源：<https://arm-software.github.io/CMSIS-DSP/latest/>

CMSIS-6 DSP – 通用格式

typedef int8_t q7_t 8-bit fractional	typedef int16x8x2_t q63x2_t vector types	typedef int16x8x2_t q15x8x2_t 16-bit fractional	typedef float32x4x4_t status64x2_t 64-bit status 12	typedef float32x4x4_t f32x4x4_t 32-bit floating	typedef int32x2x2_t q31x2x2_t 32-bit fractional	typedef int32x2x2_t q7x8x2_t 8-bit fractional	
typedef int16_t q15_t 16-bit fractional	typedef int32x4x2_t q31x4_t 32-bit fractional 12	typedef int16x8x4_t q15x8x4_t 16-bit fractional	typedef int32x4x2_t status32x4_t 32-bit status 12	typedef int32x4x2_t q31x2_t 32-bit fractional	typedef int32x4x2_t q15x8x3_t 16-bit fractional	typedef int32x4x2_t q31x2x3_t 32-bit fractional	typedef int32x4x2_t q7x8x3_t 8-bit fractional
typedef int32_t q31_t 32-bit fractional	typedef int16x8x2_t q15x8_t 16-bit fractional 12	typedef int8x16x2_t q7x16x2_t 8-bit fractional 1	typedef int16x8x2_t status16x8_t 16-bit status 12	typedef int16x8x2_t q15x4_t 16-bit fractional	typedef int8x16x3_t q7x16x3_t 8-bit fractional	typedef int32x4x2_t q31x2x4_t 32-bit fractional	typedef int8x16x4_t q7x8x4_t 8-bit fractional
typedef int64_t q63_t 64-bit fractional	typedef int8x16x2_t q7x16_t 8-bit fractional 128	typedef int8x16x4_t q7x16x4_t 8-bit fractional 1	typedef int8x16x2_t status8x16_t 8-bit status 128	typedef int8x16x2_t q7x8_t 8-bit fractional	typedef float32x2x2_t f32x2x2_t 32-bit floating	typedef float32x2x2_t q15x4x2_t 16-bit fractional	typedef float32x2x2_t status32x2_t 32-bit status 64
typedef float float32_t 32-bit floating	typedef int32x4x2_t q31x4x2_t 32-bit fractional 12	typedef int32x4x2_t q23_t 32-bit fractional	typedef float32x4_t f32x4_t 32-bit floating-p	typedef float32x2_t f32x2_t 32-bit float 6	typedef float32x2x3_t f32x2x3_t 32-bit floating	typedef float32x2x3_t q15x4x3_t 16-bit fractional	typedef float32x2x4_t status16x4_t 16-bit status 64
typedef int32x4x4_t q31x4x4_t 32-bit fractional 12	typedef int32x4x4_t q23x4_t 32-bit fractional	typedef float32x4x2_t f32x4x2_t 32-bit floating-p	typedef float32x4x3_t f32x4x3_t 32-bit floating	typedef float32x2x4_t f32x2x4_t 32-bit floating	typedef float32x2x4_t q15x4x4_t 16-bit fractional	typedef float32x2x4_t status8x8_t 8-bit status 64	

資料來源：<https://arm-software.github.io/CMSIS-DSP/latest/>

CMSIS-6 DSP – 數學函數

基本數學函式

Vector Absolute Value

Vector Addition

Vector bitwise AND

Elementwise clipping

Vector Dot Product

Vector Multiplication

Vector Negate

Vector bitwise NOT

Vector Offset

Vector bitwise inclusive OR

Vector Scale

Vector Shift

Vector Subtraction

Vector bitwise exclusive OR

快速數學函式

ArcTan2

Cosine

Fixed point division

Sine

Vector Exponential

Vector Log

Square Root

複數數學函式

Complex Conjugate

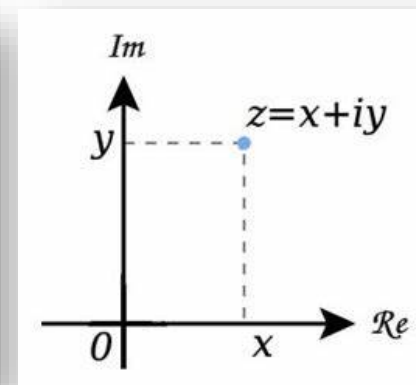
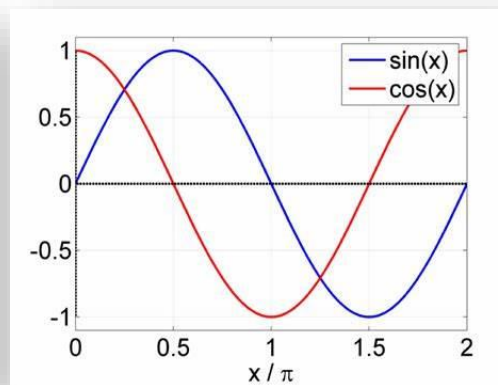
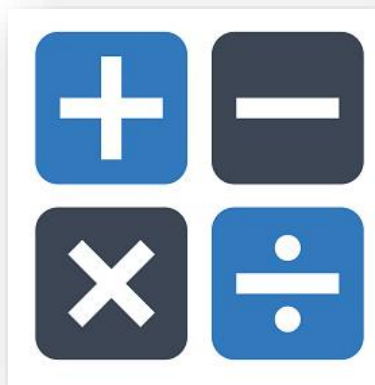
Complex Dot Product

Complex Magnitude

Complex Magnitude Squared

Complex-by-Complex Multiplication

Complex-by-Real Multiplication



資料來源：https://arm-software.github.io/CMSIS_6/latest/DSP/index.html

CMSIS-6 DSP – 濾波器函數

High Precision Q31 Biquad Cascade Filter

Biquad Cascade IIR Filters Using Direct Form I Structure

Biquad Cascade IIR Filters Using a Direct Form II Transposed Structure

Convolution

Partial Convolution

Correlation

Finite Impulse Response (FIR) Decimator

Finite Impulse Response (FIR) Filters

Finite Impulse Response (FIR) Lattice Filters

Finite Impulse Response (FIR) Sparse Filters

Infinite Impulse Response (IIR) Lattice Filters

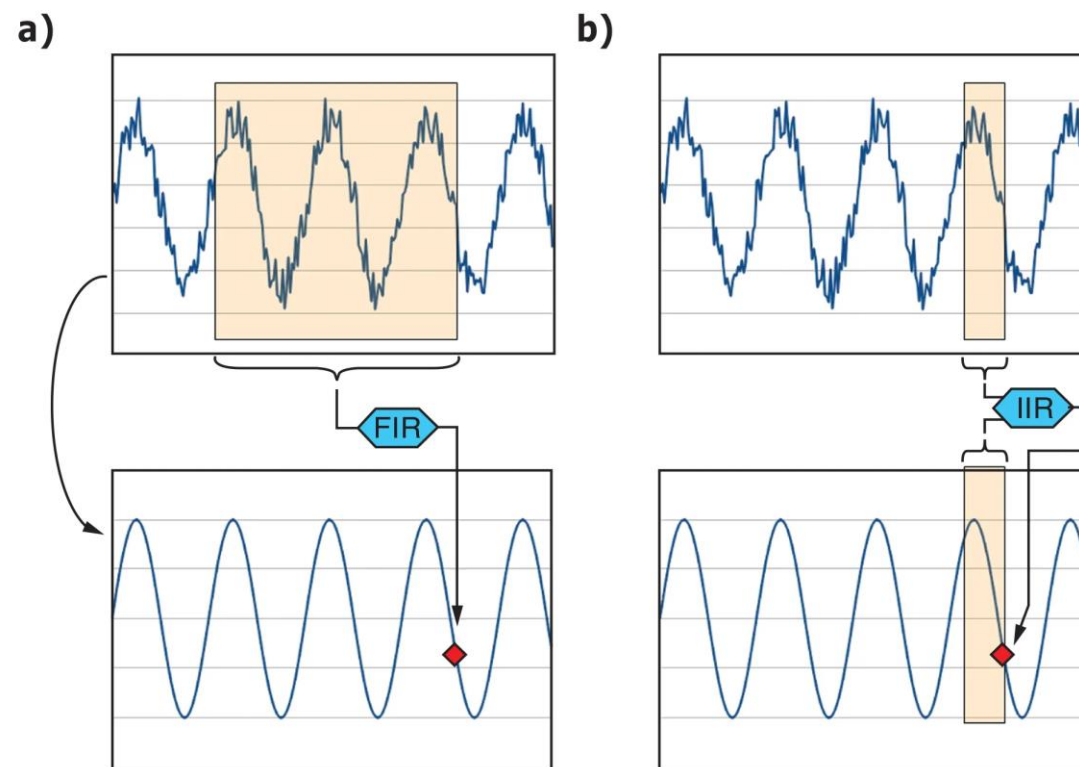
Levinson Durbin Algorithm

Least Mean Square (LMS) Filters

Normalized LMS Filters

Finite Impulse Response (FIR) Interpolator

資料來源：https://arm-software.github.io/CMSIS_6/latest/DSP/index.html



CMSIS-6 DSP – 矩陣函式

Householder transform of a vector

Matrix Addition

Cholesky and LDLT decompositions

Complex Matrix Multiplication

Complex Matrix Transpose

Matrix Initialization

Matrix Inverse

Matrix Multiplication

QR decomposition of a Matrix

Matrix Scale

Matrix Subtraction

Matrix Transpose

Matrix Vector Multiplication

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$

↕

$$\begin{aligned} c_{11} &= a_{11} * b_{11} + a_{12} * b_{21} + a_{13} * b_{31} & c_{12} &= a_{11} * b_{12} + a_{12} * b_{22} + a_{13} * b_{32} \\ c_{21} &= a_{21} * b_{11} + a_{22} * b_{21} + a_{23} * b_{31} & c_{22} &= a_{21} * b_{12} + a_{22} * b_{22} + a_{23} * b_{32} \end{aligned}$$

$$A = \begin{vmatrix} 2 & 8 & 9 \\ 1 & 9 & 7 \\ 5 & 6 & 3 \\ 0 & 4 & 2 \end{vmatrix} \quad A^T = \begin{vmatrix} 2 & 1 & 5 & 0 \\ 8 & 9 & 6 & 4 \\ 9 & 7 & 3 & 2 \end{vmatrix}$$

資料來源：https://arm-software.github.io/CMSIS_6/latest/DSP/index.html

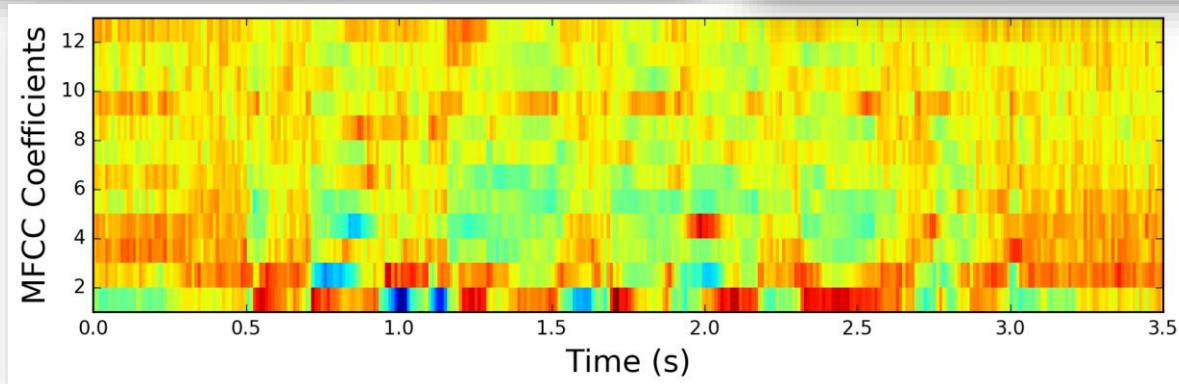
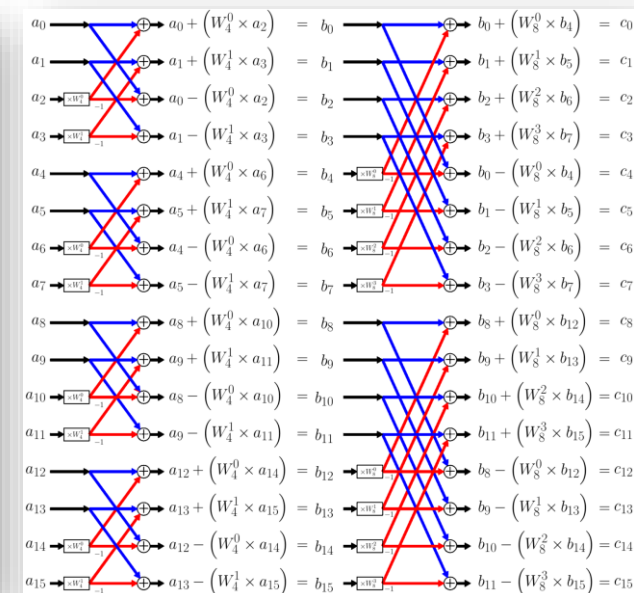
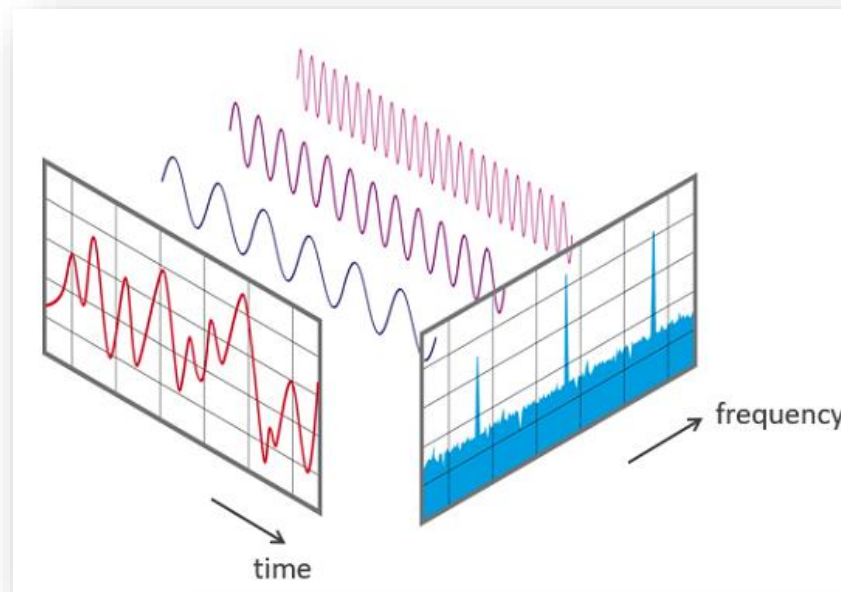
CMSIS-6 DSP – 變換函式

Complex FFT Functions

DCT Type IV Functions

MFCC

Real FFT Functions



資料來源：https://arm-software.github.io/CMSIS_6/latest/DSP/index.html

CMSIS 5 / 6 DSP 比較

主要函數差異不大，CMSIS-6 新增部份函數。

[+] New [-] Remove

Functions	CMSIS v5 (DSP v1.10.0)			v6 (DSP v1.15.0)
Basic math	Vector Absolute Value	Vector Multiplication	Vector Shift	
	Vector Addition	Vector Negate	Vector Subtraction	
	Vector bitwise AND	Vector bitwise NOT	Vector bitwise inclusive OR	
	Elementwise clipping	Vector Offset	Vector bitwise exclusive OR	
	Vector Dot Product	Vector Scale		
Bayes classifier	gaussian_naive_bayes_predict			
Complex math	Complex Conjugate	Complex Magnitude Squared		
	Complex Dot Product	Complex-by-Complex Multiplication		
	Complex Magnitude	Complex-by-Real Multiplication		
Controller	PID Motor Control	Vector Clarke Transform		
	Vector Park Transform	Vector Inverse Clarke Transform		
	Vector Inverse Park transform	Sine Cosine		
Distance	Float Distances	Boolean Distances		
Fast math	ArcTan2	Fixed point division	Vector Log	[+] Vector Exponential
	Cosine	Sine	Square Root	
Filtering	Biquad Cascade IIR Filters Using Direct Form I Structure			
	Biquad Cascade IIR Filters Using a Direct Form II Transposed Structure			
	Convolution	High Precision Q31 Biquad Cascade Filter		
	Partial Convolution	Finite Impulse Response (FIR) Decimator		
	Correlation	Finite Impulse Response (FIR) Filters		
	Levinson Durbin Algorithm	Finite Impulse Response (FIR) Lattice Filters		
	Least Mean Square (LMS) Filters	Finite Impulse Response (FIR) Sparse Filters		
	Normalized LMS Filters	Infinite Impulse Response (IIR) Lattice Filters		
		Finite Impulse Response (FIR) Interpolator		

Interpolation	Bilinear Interpolation	Linear Interpolation	Cubic Spline Interpolation	
Matrix	Matrix Addition	Matrix Scale	Cholesky and LDLT decompositions	[+] Householder transform of a vector [+] QR decomposition of a Matrix
	Matrix Initialization	Matrix Subtraction	Complex Matrix Multiplication	
	Matrix Inverse	Matrix Transpose	Complex Matrix Transpose	
	Matrix Multiplication		Matrix Vector Multiplication	
Quaternion	Quaternion conversions	Quaternion Inverse	Quaternion normalization	
	Quaternion Conjugate	Quaternion Norm	Quaternion Product	
Statistical	Absolute Maximum	Mean	Mean Square Error	[+] Accumulation functions
	Absolute Minimum	Minimum	Kullback-Leibler divergence	
	Entropy	Power	Root mean square (RMS)	
	LogSumExp	Variance	Standard deviation	
	Maximum			
Support	Typecasting	Weighted Sum	Convert 16-bit fixed point value	[+] Convert 64-bit floating point value
	Barycenter	Vector sorting algorithms	Convert 32-bit fixed point value	
	Vector Copy	Convert 16-bit floating point value	Convert 8-bit fixed point value	
	Vector Fill	Convert 32-bit floating point value		
SVM	Linear SVM	RBF SVM		
	Polynomial SVM	Sigmoid SVM		
Transform	Complex FFT Functions		MFCC	
	DCT Type IV Functions		Real FFT Functions	
Window	none			[+] Flat-top window [+] Regular windos

Vector Clarke Transform

Vector Inverse Clarke Transform

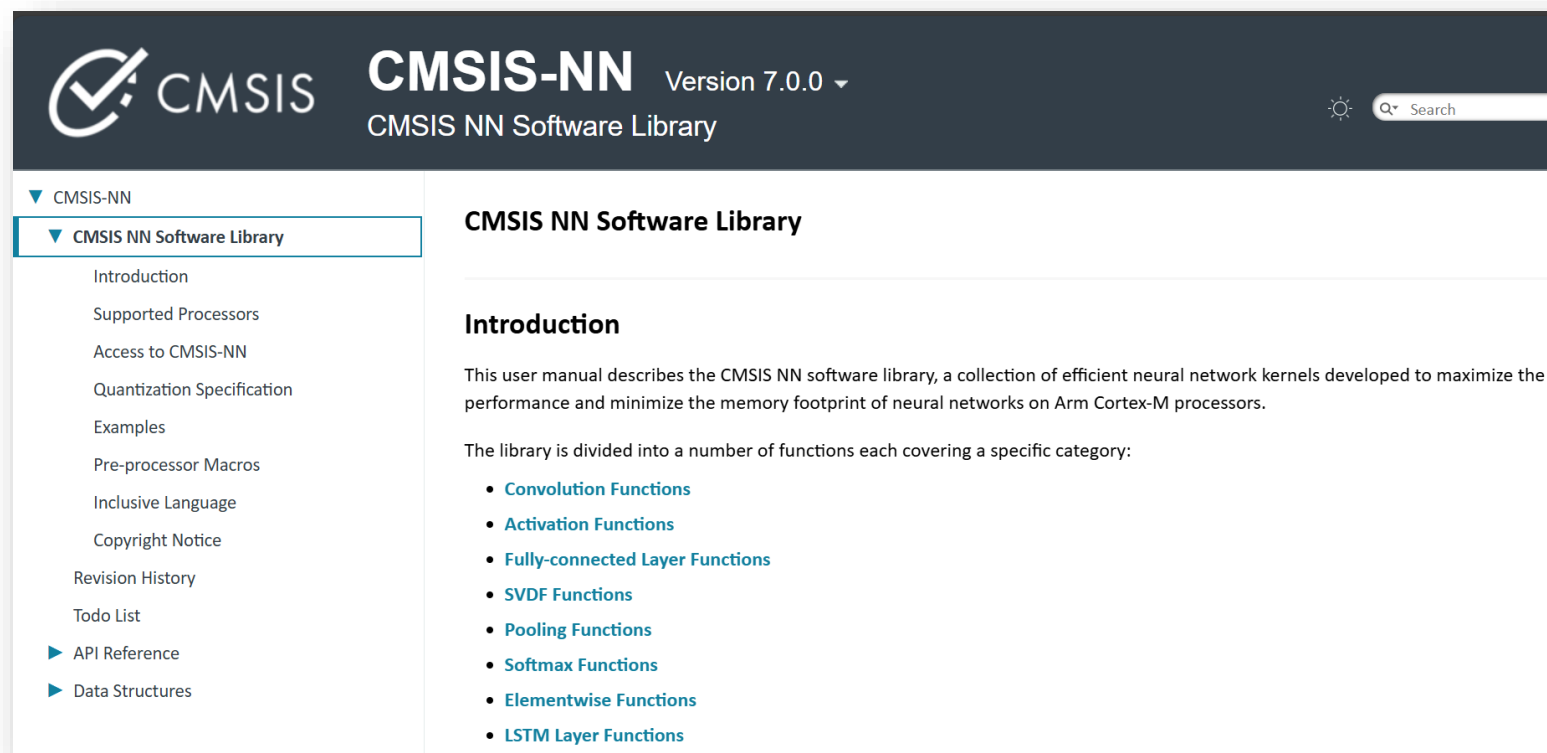
Sine Cosine

OmniXRI整理製作, 2024/02/16

資料來源：<https://omnixri.blogspot.com/2024/02/tinyml-arm-cmsis-6-dsp-nn.html>

CMSIS-6 NN

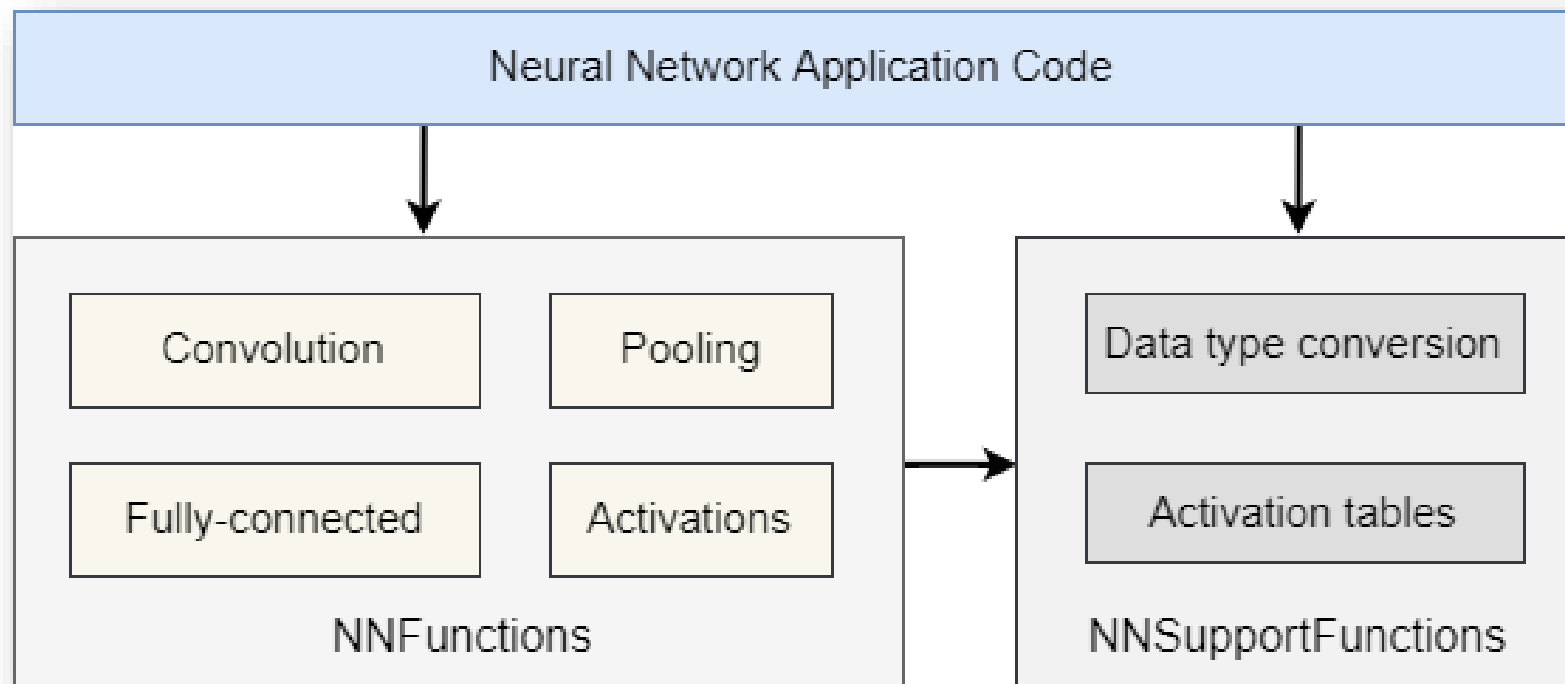
CMSIS-NN 是一個開源軟體庫（在Github已獨立），它提供了一組高效的神經網路（NN）內核，這些內核旨在最大限度地提高在 Arm Cortex-M 處理器上運行的神經網路的性能並最大限度地減少記憶體佔用。



- 卷積函數
- 激勵函數
- 全連結層函數
- SVDF函數
- 池化函數
- Softmax函數
- 元素級別函式
- LSTM函數

資料來源：<https://arm-software.github.io/CMSIS-NN/latest/>

CMSIS-6 NN 架構圖



CMSIS-NN 可支援

- 沒有SIMD指令的處理器，如 Cortex-M0/0+
- 帶有DSP擴展指令的處理器，如 Cortex-M4/M7
- 帶有MVE指令的處理器，如 Cortex-M55/M85
- **不直接支援 MicroNPU Ehtos-U55/U65/U85**
- **不使用 CMSIS-NN 也可以使用傳統指令集執行相關計算，只是速度較慢。**

資料來源：<https://arm-software.github.io/CMSIS-NN/latest/>

CMSIS-6 NN – 卷積函式

arm_convolve_XXX

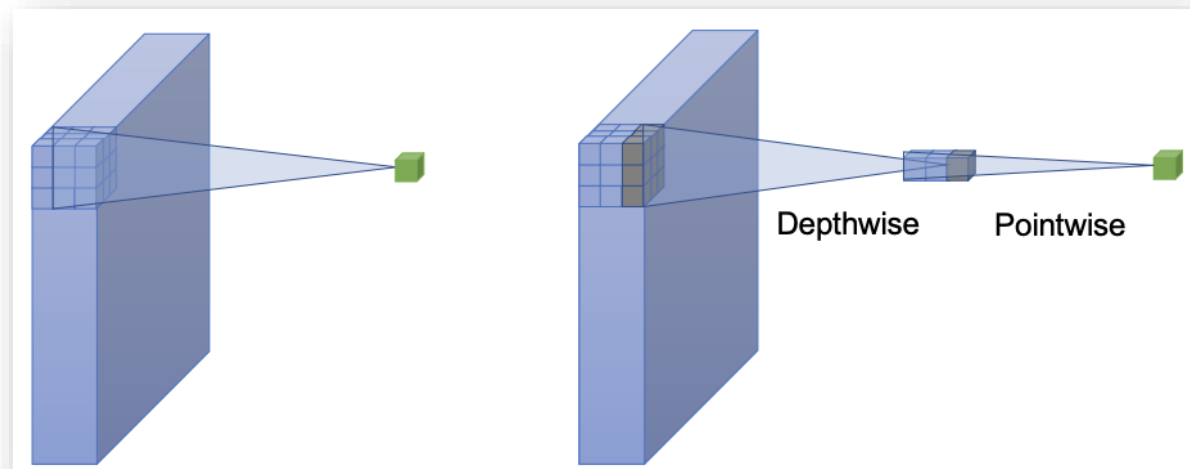
- 1_x_n_s4, 1_x_n_s8, 1x1_s4, 1x1_s4_fast, 1x1_s8, 1x1_s8_fast, even_s4, s16, s4, s8, wrapper_s16, wrapper_s4, wrapper_s8

arm_depthwise_conv_XXX

- 3x3_s8, fast_s16, s16, s4, s4_opt, s8, s8_opt, wrapper_s16, wrapper_s4, wrapper_s8

arm_transpose_conv_XXX

- s8, wrapper_s8



資料來源：<https://arm-software.github.io/CMSIS-NN/latest/>

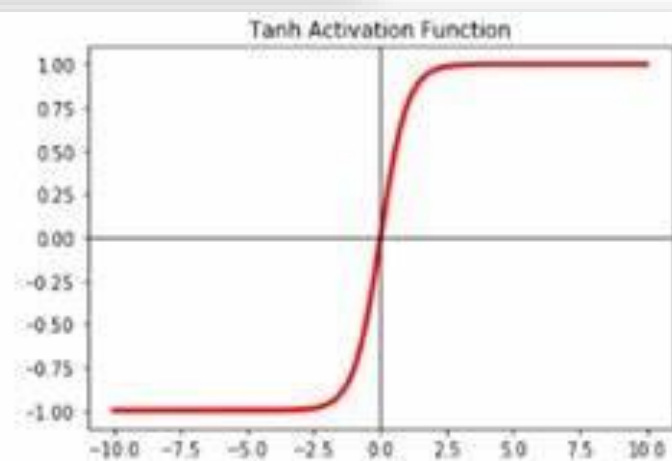
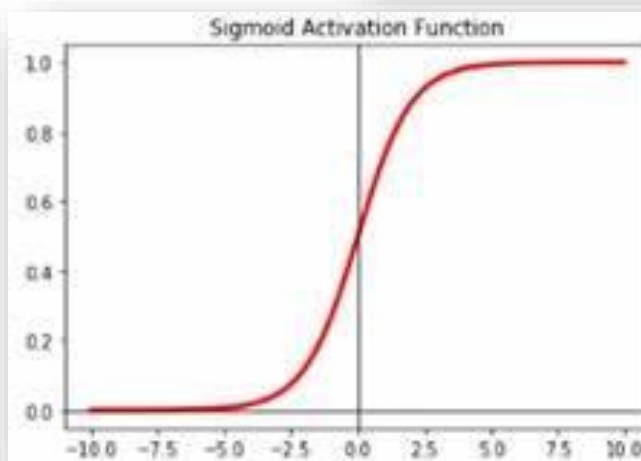
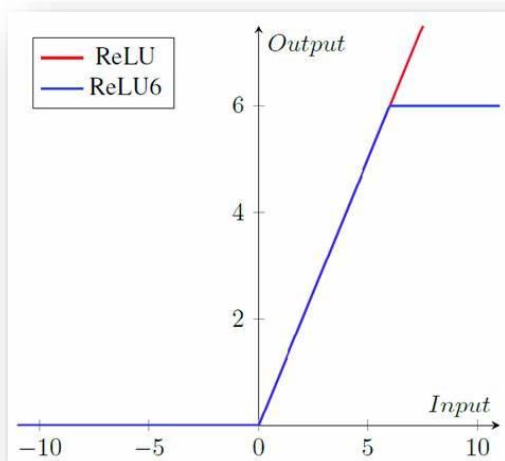
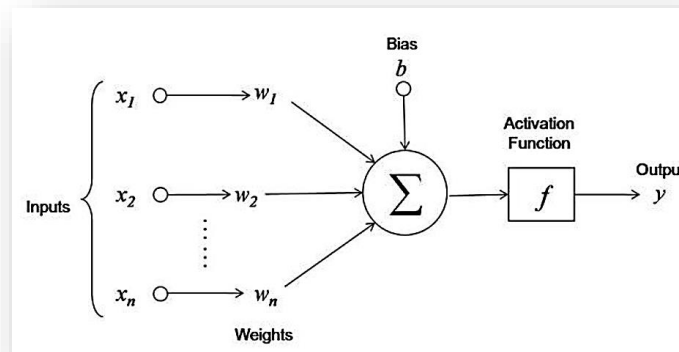
CMSIS-6 NN – 激勵函式

arm_nn_activation_s16 (ARM_SIGMOID, ARM_TANH)

arm_relu6_s8

arm_relu_q15

arm_relu_q7



資料來源：<https://arm-software.github.io/CMSIS-NN/latest/>

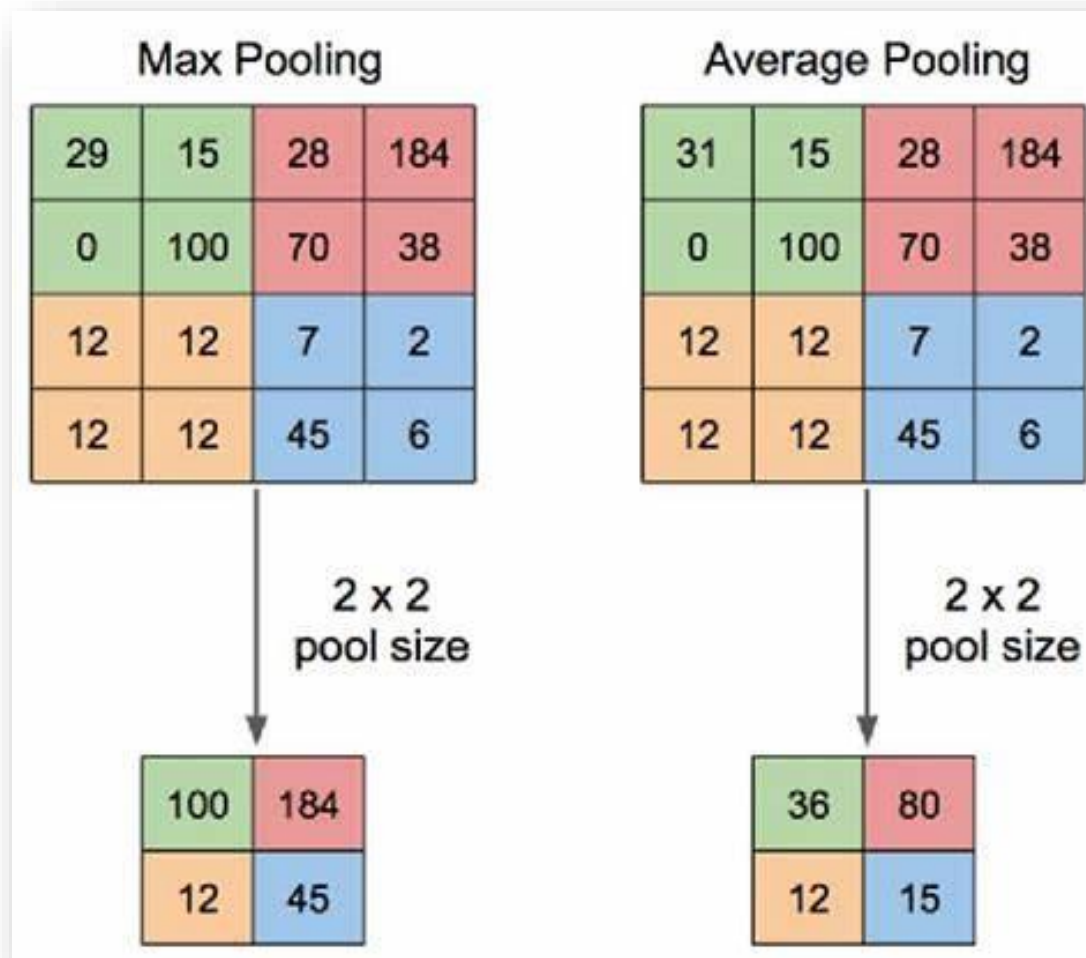
CMSIS-6 NN – 池化函式

arm_avgpool_s16

arm_avgpool_s8

arm_max_pool_s16

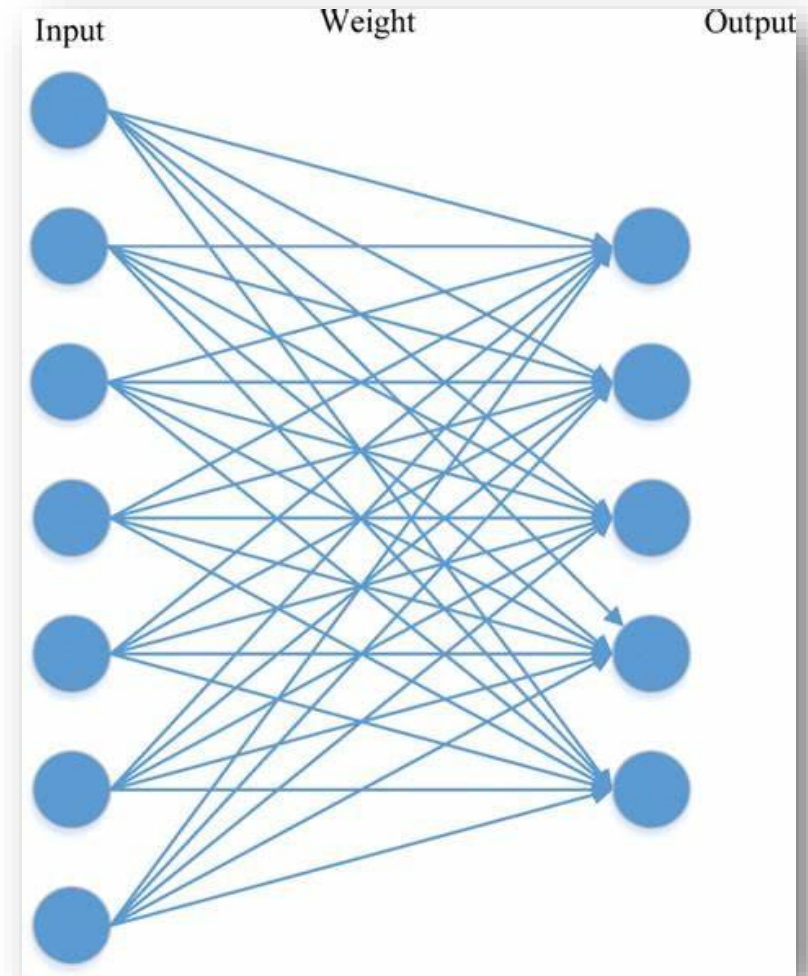
arm_max_pool_s8



資料來源：<https://arm-software.github.io/CMSIS-NN/latest/>

CMSIS-6 NN – 全連結層函式

arm_batch_matmul_s16
arm_batch_matmul_s8
arm_fully_connected_per_channel_s8
arm_fully_connected_s16
arm_fully_connected_s4
arm_fully_connected_s8
arm_fully_connected_wrapper_s8
arm_vector_sum_s8
arm_vector_sum_s8_s64



資料來源：<https://arm-software.github.io/CMSIS-NN/latest/>

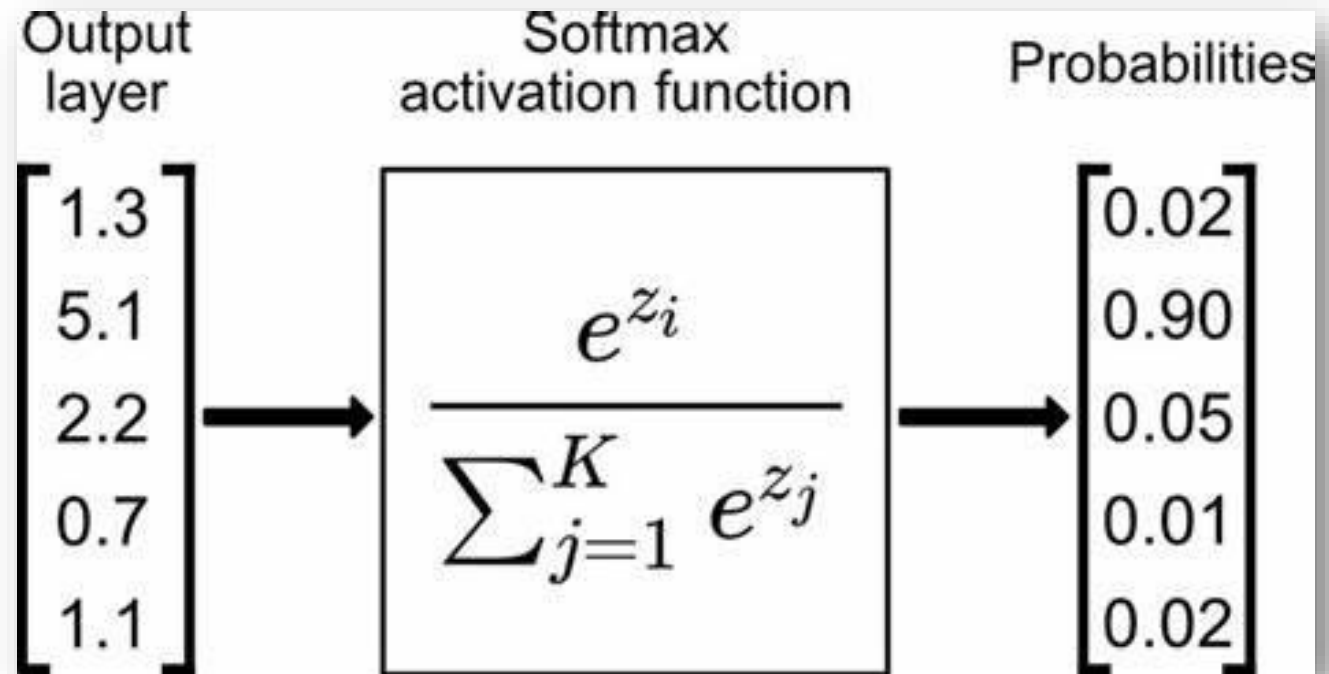
CMSIS-6 NN – Softmax函式

arm_softmax_s16

arm_softmax_s8

arm_softmax_s8_s16

arm_softmax_u8



資料來源：<https://arm-software.github.io/CMSIS-NN/latest/>

CMSIS 5 / 6 NN 比較

[+] New [-] Remove [*] private

Functions	CMSIS v5 (NN v3.1.0)	v6 (NN 5.0.0)
Activation	arm_nn_activations_direct_q15 arm_nn_activations_direct_q7	arm_relu6_s8 arm_relu_q15 arm_relu_q7 [+] ..._activation_s16
Basic math	arm_elementwise_add_s16 arm_elementwise_add_s8	arm_elementwise_mul_s16 arm_elementwise_mul_s8 Rename [Elementwise]
Concatenation	arm_concatenation_s8_w arm_concatenation_s8_x	arm_concatenation_s8_y arm_concatenation_s8_z
Fully-connected	arm_fully_connected_mat_q7_vec_q15 arm_fully_connected_q15 arm_fully_connected_q7 arm_fully_connected_s16 arm_fully_connected_s8	arm_fully_connected_mat_q7_vec_q15_opt arm_fully_connected_q15_opt arm_fully_connected_q7_opt arm_fully_connected_s16_get_buffer_size arm_fully_connected_s8_get_buffer_size [+] arm_fully_connected_s4 [+] arm_vector_sum_s8
LSTM Layer	None	[+] arm_lstm_unidirectional_s16_s8
Pooling	arm_avgpool_s16 arm_avgpool_s8 arm_max_pool_s16 arm_maxpool_q7_HWC	arm_avgpool_s16_get_buffer_size arm_avgpool_s8_get_buffer_size arm_max_pool_s8 arm_avepool_q7_HWC
Reshape	arm_reshape_s8	
Softmax	arm_nn_softmax_common_s8 arm_softmax_q15 arm_softmax_q7 arm_softmax_s16	arm_softmax_s8 arm_softmax_s8_s16 arm_softmax_u8 arm_softmax_with_batch_q7 [*] arm_nn_softmax_common_s8
SVDF	arm_svdf_s8	arm_svdf_state_s16_s8
Conversion	arm_q7_to_q15_no_shift arm_q7_to_q15_reordered_no_shift arm_nn_accumulate_q7_to_q15 arm_nn_add_q7	arm_q7_to_q15_reordered_with_offset arm_q7_to_q15_with_offset arm_nn_mat_mult_nt_t_s8 arm_nn_mult_q15
Basic Math	arm_nn_depthwise_conv_nt_t_padded_s8 arm_nn_depthwise_conv_nt_t_s8 arm_nn_mat_mul_core_1x_s8 arm_nn_mat_mul_core_4x_s8	arm_nn_mult_q7 arm_nn_vec_mat_mult_t_s16 arm_nn_vec_mat_mult_t_s8 arm_nn_vec_mat_mult_t_svdf_s8 [*] arm_elementwise_mul_s16_s8

OmniXRI整理製作, 2024/02/16

差異頗大，許多函式被刪除或整併。

[+] New [-] Remove [*] private

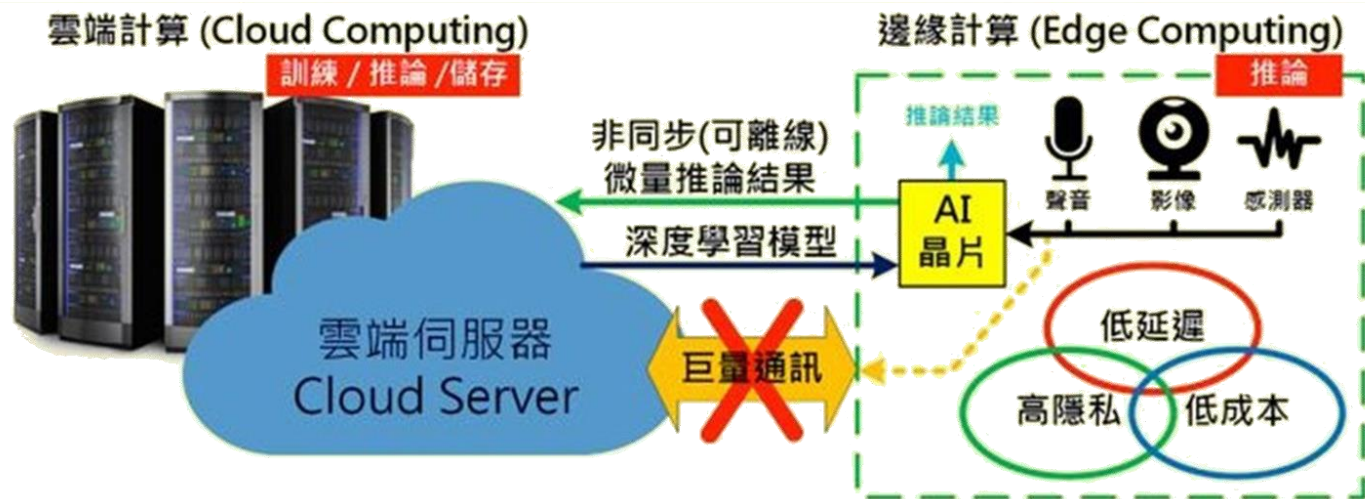
Functions	CMSIS v5 (NN v3.1.0)	v6 (NN 5.0.0)
Convolution	arm_convolve_1_x_n_s8 arm_convolve_1_x_n_s8_get_buffer_size arm_convolve_1x1_HWC_q7_fast_nonsquare arm_convolve_1x1_s8_fast arm_convolve_1x1_s8_fast_get_buffer_size arm_convolve_fast_s16 arm_convolve_fast_s16_get_buffer_size arm_convolve_HWC_q15_basic arm_convolve_HWC_q15_fast arm_convolve_HWC_q15_fast_nonsquare arm_convolve_HWC_q7_basic arm_convolve_HWC_q7_basic_nonsquare arm_convolve_HWC_q7_fast arm_convolve_HWC_q7_fast_nonsquare arm_convolve_HWC_q7_RGB arm_convolve_s16 arm_convolve_s16_get_buffer_size arm_convolve_s8 arm_convolve_s8_get_buffer_size arm_convolve_wrapper_s16 arm_convolve_wrapper_s16_get_buffer_size arm_convolve_wrapper_s8 arm_convolve_wrapper_s8_get_buffer_size arm_depthwise_conv_3x3_s8 depthwise_conv_s16_generic_s16 arm_depthwise_conv_s16 depthwise_conv_s8_mult_4 depthwise_conv_s8_generic arm_depthwise_conv_s8 arm_depthwise_conv_s8_opt arm_depthwise_conv_s8_opt_get_buffer_size depthwise_conv_u8_mult_4 depthwise_conv_u8_generic arm_depthwise_conv_u8_basic_ver1 arm_depthwise_conv_wrapper_s8 arm_depthwise_conv_wrapper_s8_get_buffer_size arm_depthwise_separable_conv_HWC_q7 arm_depthwise_separable_conv_HWC_q7_nonsquare	arm_convolve_1_x_n_s8 [+] arm_convolve_1x1_s4 [+] arm_convolve_1x1_s4_fast [+] arm_convolve_1x1_s8 arm_convolve_1x1_s8_fast arm_convolve_fast_s16 arm_convolve_s16 [+] arm_convolve_s4 arm_convolve_s8 arm_convolve_wrapper_s16 [+] arm_convolve_wrapper_s4 arm_convolve_wrapper_s8 arm_depthwise_conv_3x3_s8 [+] arm_depthwise_conv_fast_s16 arm_depthwise_conv_s16 [+] arm_depthwise_conv_s4 [+] arm_depthwise_conv_s4_opt arm_depthwise_conv_s8 arm_depthwise_conv_s8_opt [+] arm_depthwise_conv_wrapper_s16 [+] arm_depthwise_conv_wrapper_s4 arm_depthwise_conv_wrapper_s8 [+] arm_transpose_conv_s8 V5 Basic Math move to V6 Convolution Private [*] arm_nn_depthwise_conv_nt_t_padded_s8 [*] arm_nn_depthwise_conv_nt_t_s8 [*] arm_nn_mat_mult_nt_t_s8 [*] arm_nn_mat_mul_core_1x_s8 [*] arm_nn_mat_mul_core_4x_s8

OmniXRI整理製作, 2024/02/16

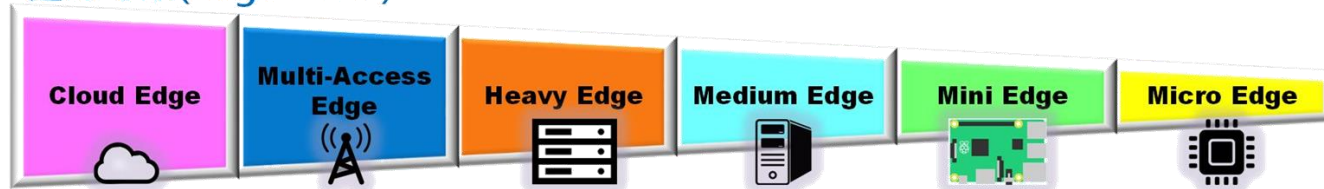
資料來源：<https://omnixri.blogspot.com/2024/02/tinyml-arm-cmsis-6-dsp-nn.html>



Edge AI

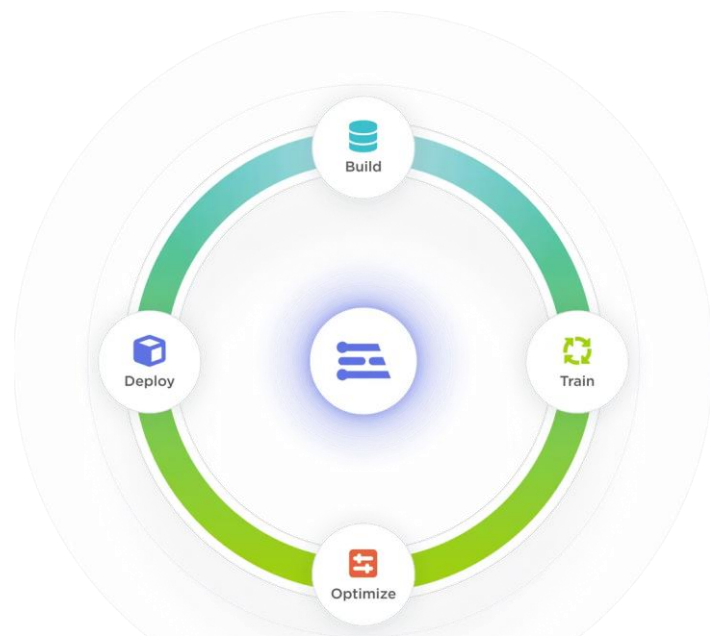


邊緣等級(Edge Level)



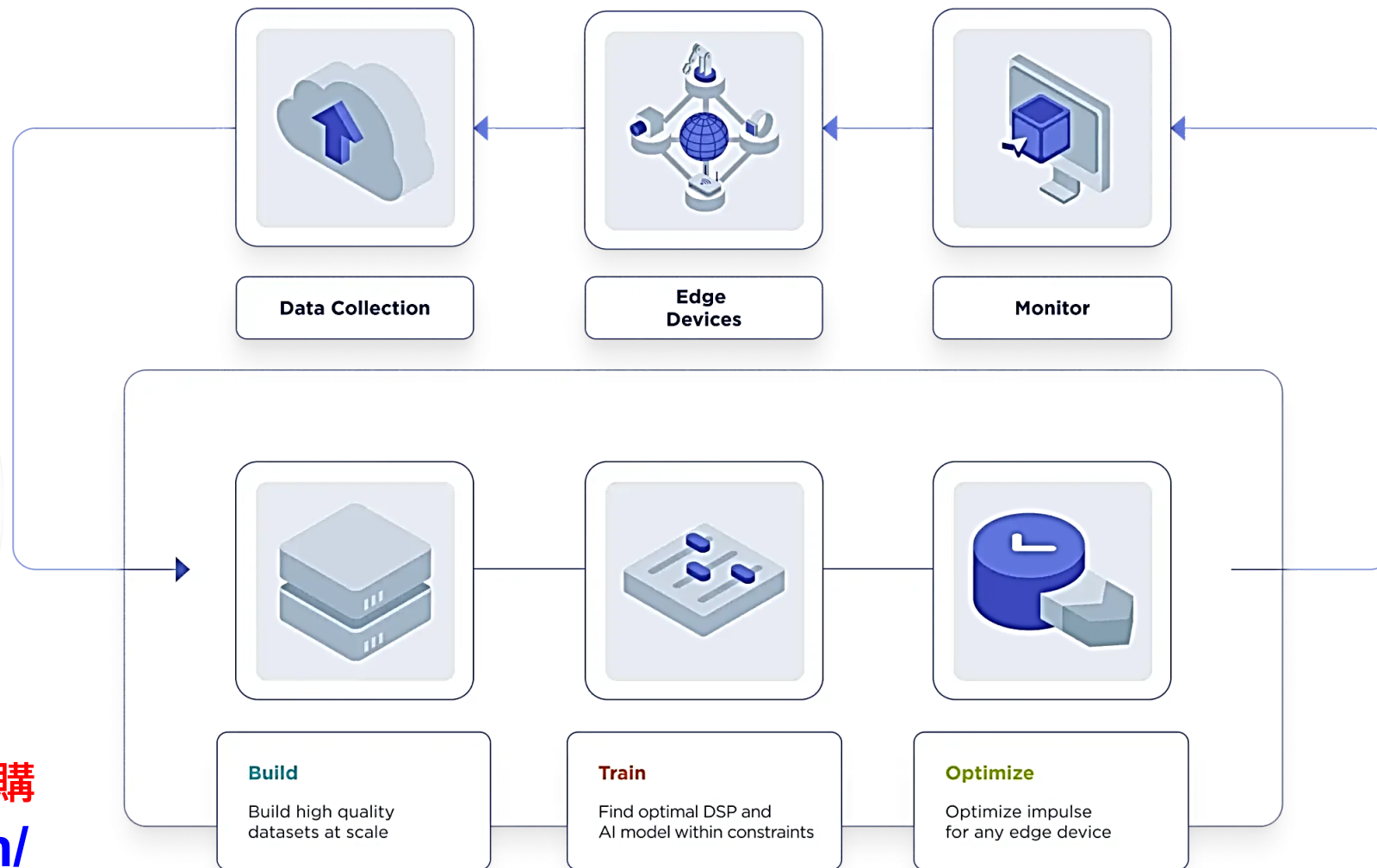
4.2. EDGE IMPULSE

Edge Impulse 簡介



2019成立，2025/3 已被高通收購

<https://edgeimpulse.com/>



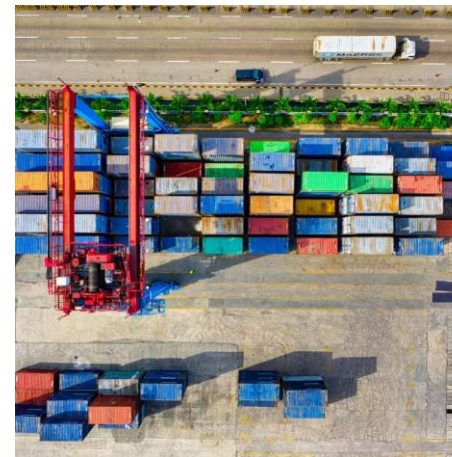
Edge Impulse 應用實例



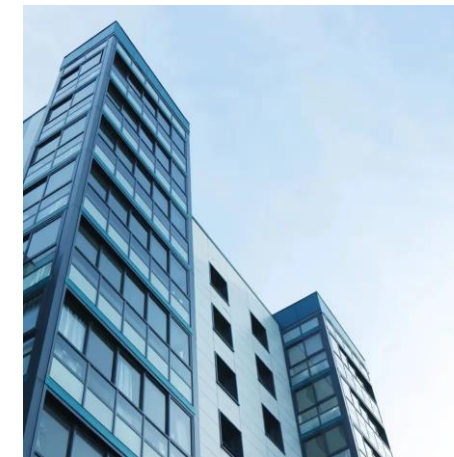
智慧家庭



智慧醫療



基礎建設



智慧建築



智慧工廠



智慧穿戴

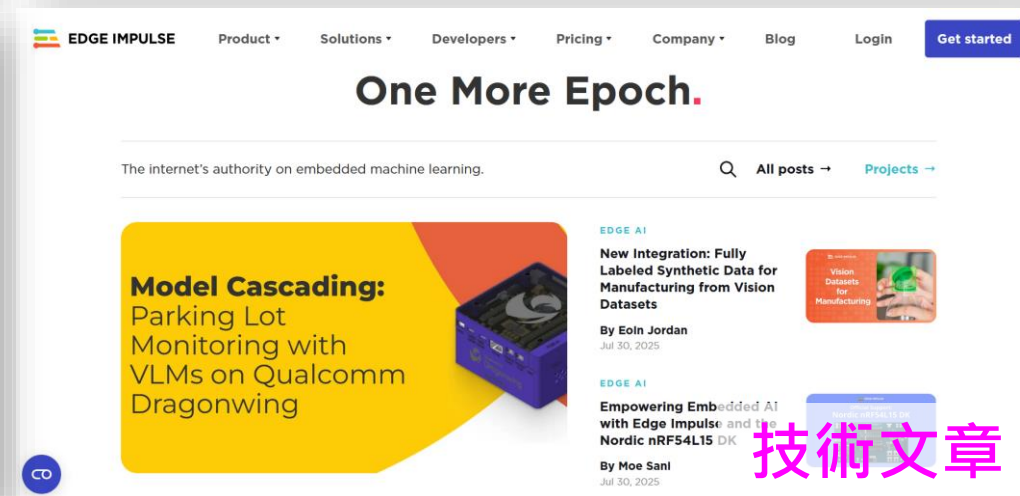
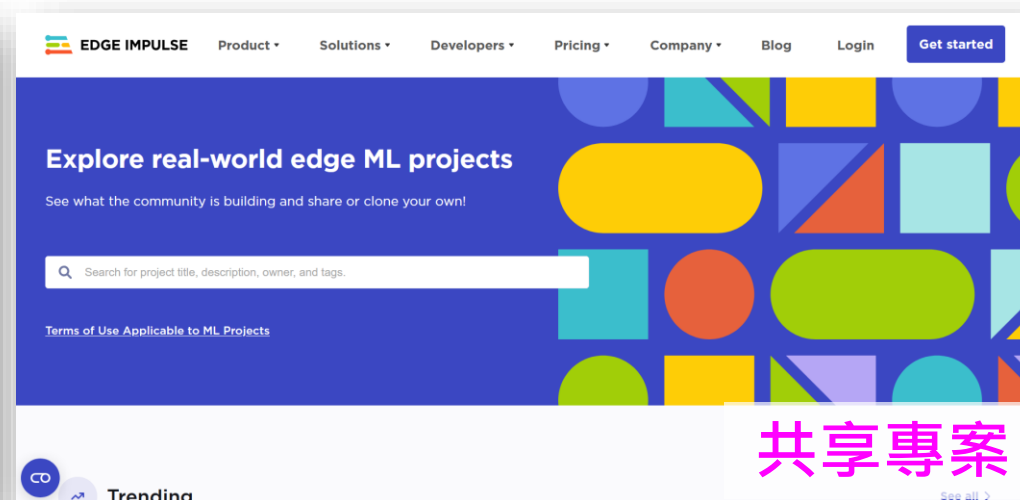
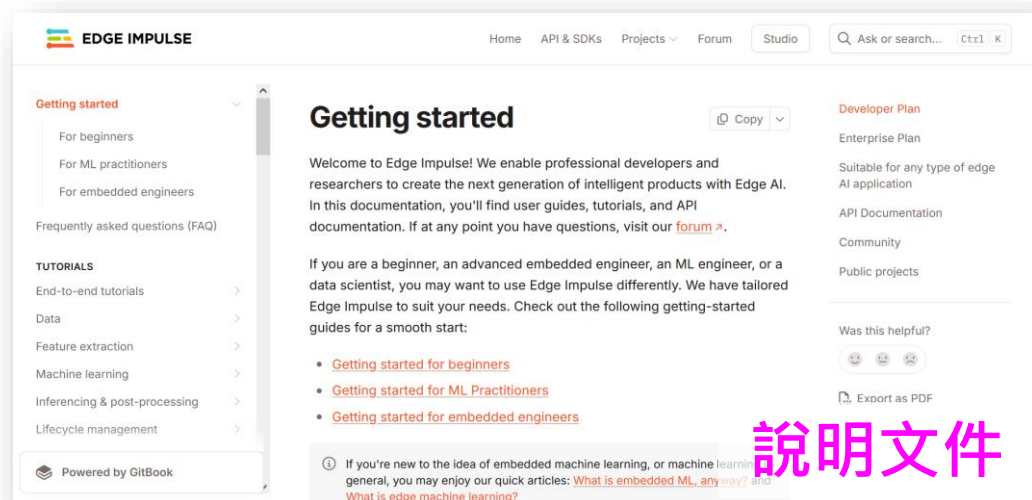


資產追蹤



人機互動

Edge Impulse 技術支援



Arduino UNO Q / Ventuno Q

Arduino UNO Q



Qualcomm



Edge Impulse 2025/3 已被高通收購
Arduino 2025/10 已被高通收購

- Qualcomm Dragonwing QRB2210
 - Arm Cortex-A53 @2GHz
 - Adreno 702 GPU @845MHz
 - DSP x2
- STM32U585(Cortex-M33@160Mhz)
- 2G/4GB LPDDR4, 16GB eMMC
- Linux Debian, Arduino App Lab

<https://www.arduino.cc/product-uno-q>



Arduino
App
Lab

<https://docs.arduino.cc/software/app-lab/>

Arduino Ventuno Q



- Qualcomm Dragonwing IQ8275
 - Kryo™ Gen 6 CPU
 - Adreno 623 GPU
 - NPU 40TOPS
- STM32H5(Cortex-M33@250MHz)
- 16GB LPDDR4, 64GB eMMC
- Linux Debian, Arduino App Lab

<https://www.arduino.cc/product-ventuno-q/>

Edge Impulse 說明文件 2025 更新版

The screenshot shows the Edge Impulse documentation website. The navigation bar includes 'Knowledge', 'Studio', 'Hardware', 'Tools', 'APIs', 'Tutorials', 'Projects', and 'Datasets'. The 'Knowledge' section is highlighted with a blue box. Below it, several sub-sections are visible: 'Studio', 'Hardware', 'Tools', 'APIs', 'Tutorials', 'Projects', and 'Datasets'. The 'Studio' section has an orange box around the 'Projects' icon and the text '工作專案' (Work Projects). The 'Tools' section has a red box around the 'End-to-end' icon. The 'Tutorials' section has a red box around the 'Image tutorials' icon and the text '專家網路 端到端教學 影像、聲音、時序 (感測器)、指定硬體' (Expert network End-to-end teaching Image, Audio, Time-series (Sensors), Specified hardware). The 'Projects' section has a green box around the 'Expert network' icon.

工作專案

專家網路 端到端教學 影像、聲音、時序 (感測器)、指定硬體

<https://docs.edgeimpulse.com/>

Edge Impulse 支援硬體及推論工具

EDGE AI HARDWARE

Overview

Production-ready

MCU

MCU + AI accelerators

CPU

CPU + AI accelerators

GPU

Mobile phone

Porting guide

自定義
開發板

RUN INFERENCE

C++ library

Arm Keil MDK CMSIS-PACK

Arduino library

Cube.MX CMSIS-PACK

Docker container

DRP-AI library

IAR library

Linux EIM executable

OpenMV

Particle library

Qualcomm IM SDK GStreamer

Qualcomm Device Cloud

WebAssembly

Edge Impulse firmwares

Hardware specific tutorials

資料來源：<https://docs.edgeimpulse.com/docs/tools/edge-impulse-python-sdk>

Edge Impulse Studio — 專案功能表

EDGE IMPULSE

 Dashboard

 Devices

 Data acquisition

 Impulse design

 Create impulse

 Spectrogram

 NN Classifier

專案儀錶板

連線裝置

資料擷取

流程設計

建立流程

資料頻譜

神經網路分類器

不同類型專案項目不同

 EON Tuner

 Retrain model

 Live classification

 Model testing

 Performance calibration

 Versioning

 Deployment

EON調整器

重新訓練模型

立即分類

模型測試

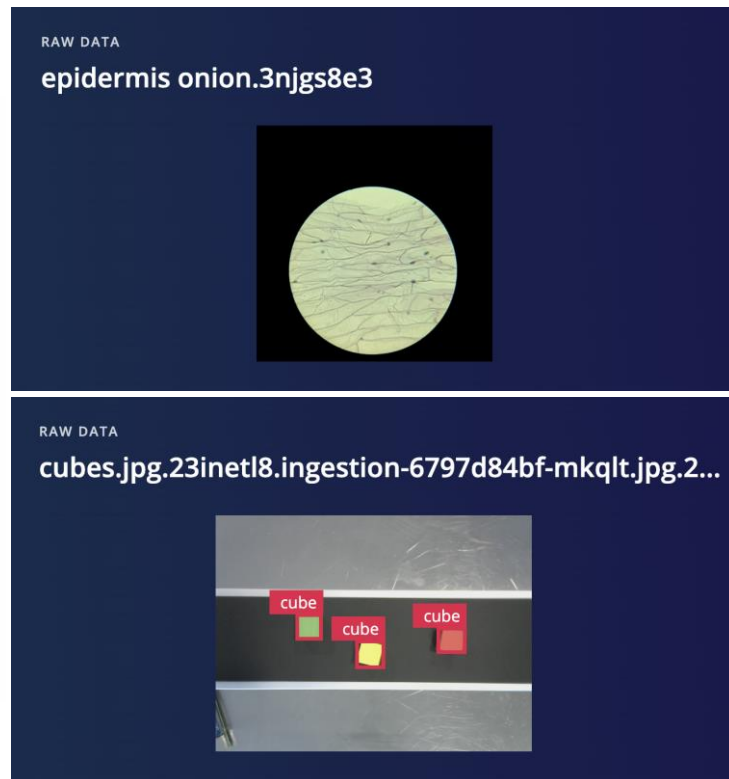
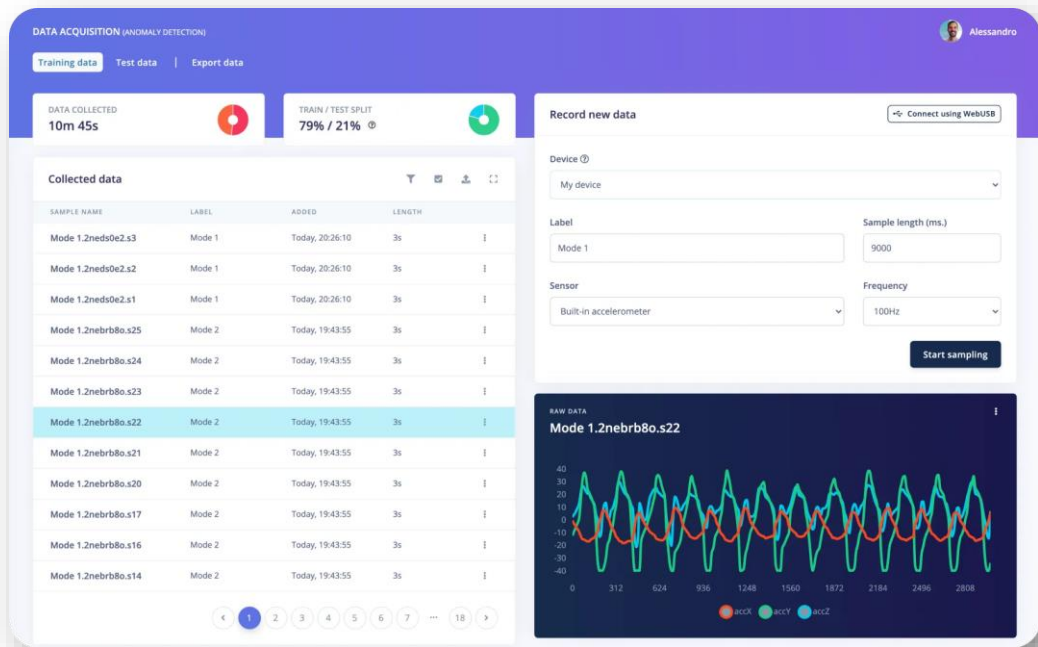
效能校正

版本管理

程式部署

資料來源：<https://docs.edgeimpulse.com/studio>

Edge Impulse 主要功能 — 資料集建置



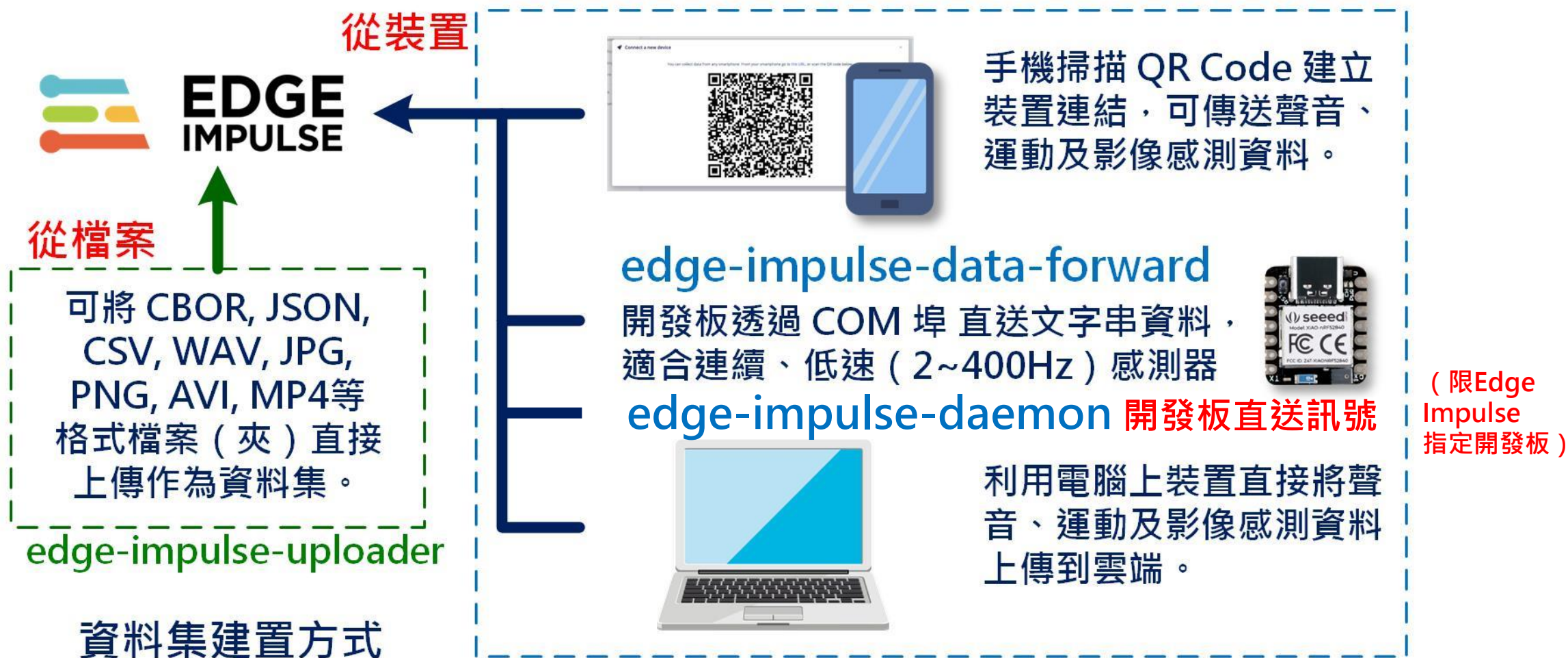
- 外部資料上傳
- 外部感測上傳
- 連續影像標註
- AI自動標註

多維感測器時序信號手動及自動分割

整張影像及物件標註

資料來源：<https://docs.edgeimpulse.com/studio/projects/data-acquisition>

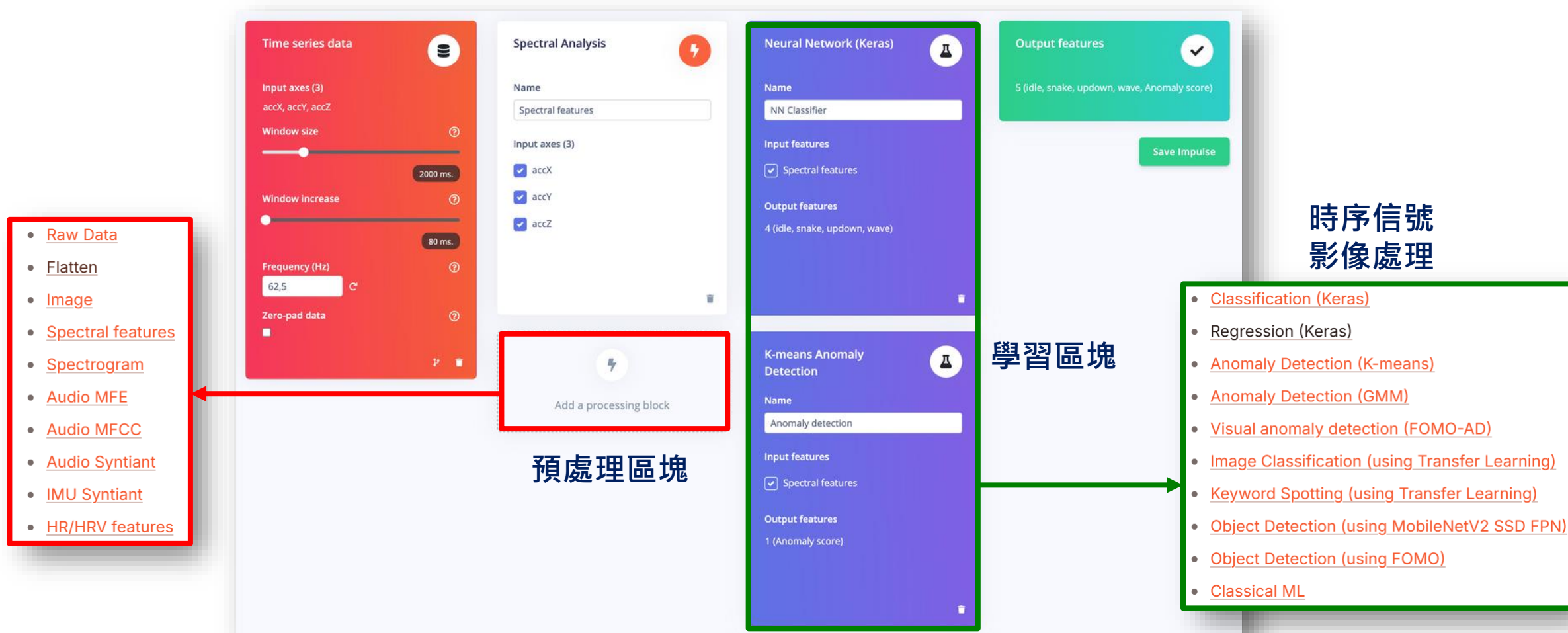
Edge Impulse 資料集建置方式



資料來源：<https://docs.edgeimpulse.com/studio/projects/data-acquisition>

OmniXRI整理製作, 2024/05/09

Edge Impulse 主要功能 — 模型設定

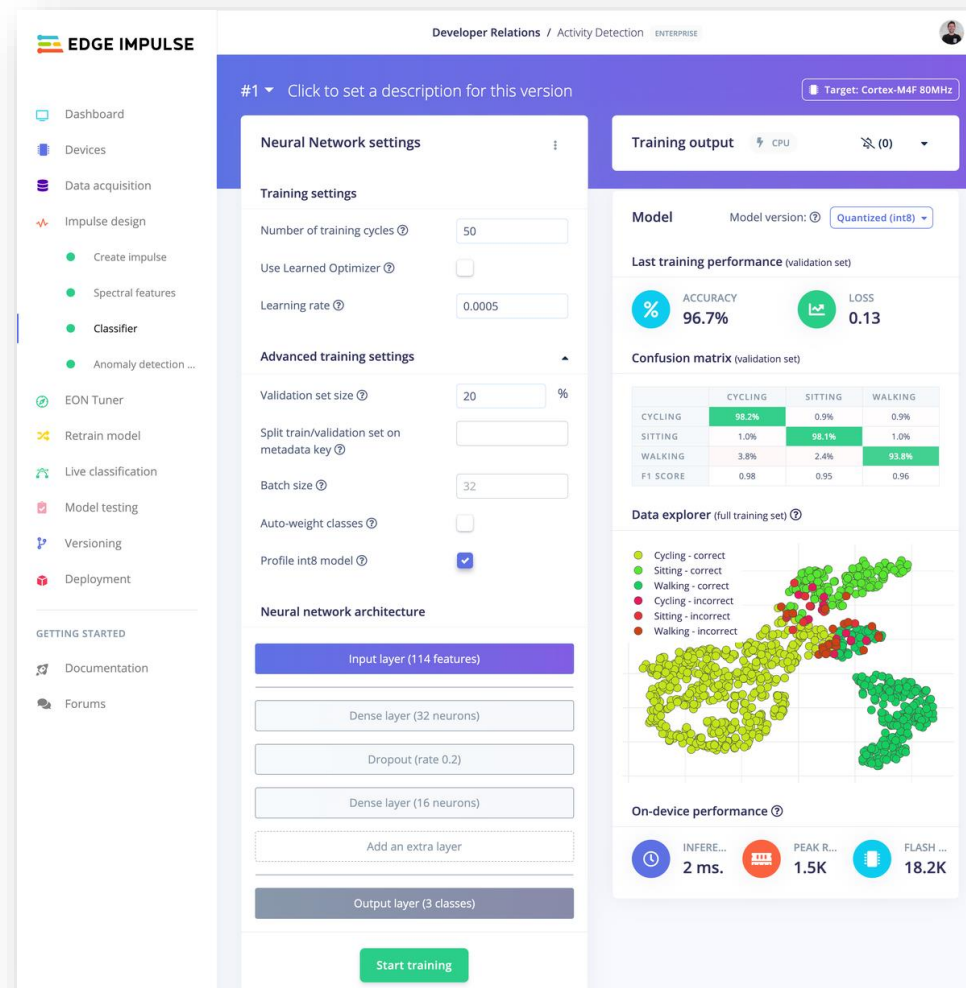


資料來源：<https://docs.edgeimpulse.com/studio/projects/impulse-design>

Edge Impulse 主要功能 — 模型訓練

訓練超參數

- 訓練次數
- 優化器
- 學習率
- 批次大小
- 模型結構
(可自定義)

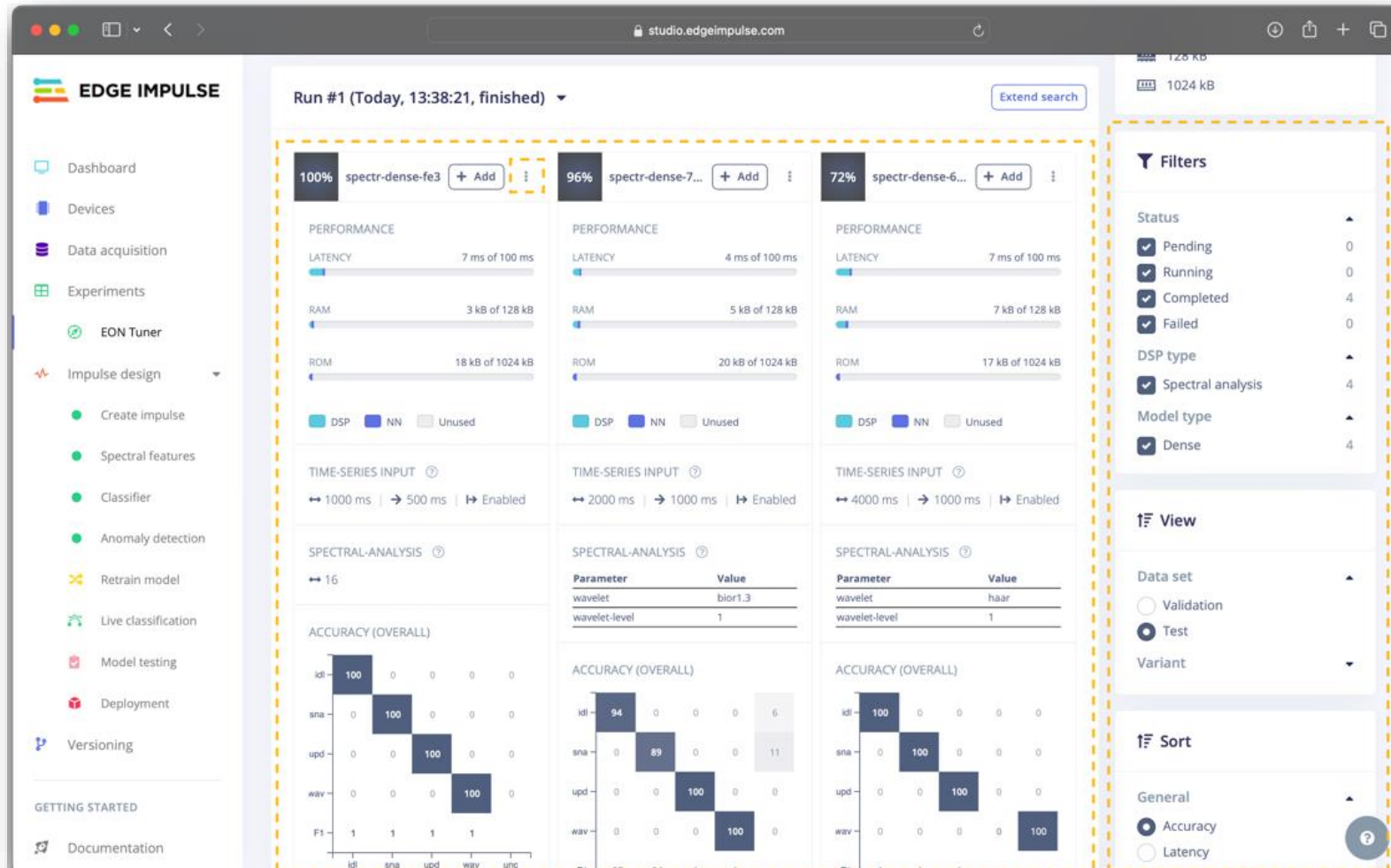


訓練結果

- 推論精度 / 損失率
- 混淆矩陣
- 資料分佈
- FLASH / SRAM
使用量
- 推論預估時間

資料來源：<https://docs.edgeimpulse.com/studio/projects/learning-blocks/training-graphs>

Edge Impulse主要功能—EON 模型優化



- Flash 優先
- SRAM 優先
- 推論精度優先
- 推論速度（延遲）優先

自動產生多組結果，非常耗時，為選配性項目。

資料來源：<https://docs.edgeimpulse.com/studio/projects/eon-tuner>

Edge Impulse 主要功能 — 部署

Configure your deployment 部署方式

You can deploy your impulse to any device. This makes the model run without an internet connection, minimizes latency, and runs with minimal power consumption. [Read more.](#)

Search deployment options

- C++ library**
A portable C++ library with no external dependencies, which can be compiled with any modern C++ compiler.
- Arduino library**
An Arduino library with examples that runs on most Arm-based Arduino development boards.
- Cube.MX CMSIS-PACK**
A STM32Cube.MX CMSIS-PACK, for fast inferencing on many ST MCUs.
- Ethos-U library**
A C++ library with inferencing for devices with an Ethos-U NPU.
- Meta TF Model**
A Meta TF converted saved model (.flat) format with the AI deployment...

MODEL OPTIMIZATIONS 部署優化

Model optimizations can increase on-device performance but may reduce accuracy.

EON™ Compiler (RAM optimized) ENTERPRISE

Same accuracy, 49% less RAM, 11% less ROM.

Quantized (int8) Selected ✓

	IMAGE	TRANSFER LE...	TOTAL
LATENCY	4 ms.	187 ms.	191 ms.
RAM	4.0K	416.1K	416.1K
FLASH	-	701.3K	-
ACCURACY			97.30%

Unoptimized (float32) Select

	IMAGE	TRANSFER LE...	TOTAL
LATENCY	4 ms.	307 ms.	311 ms.
RAM	4.0K	1.1M	1.1M
FLASH	-	2.0M	-
ACCURACY			97.30%

Estimate for Cortex-M7 216MHz - [Change target](#)

資料來源：<https://docs.edgeimpulse.com/studio/projects/deployment>

Edge Impulse - 註冊/登入帳號

Gmail
直接登入

Sign up to start building your edge AI models in minutes!

Continue with Google

Continue with GitHub

or

Email

Enter an email

電子信箱

Username

Enter a username

使用者姓名

Password

Create a password

密碼

What describes your role at your company? (optional)

Select role

Let us know you're human

成功!

☐ I accept the Privacy Policy, Developer Terms of Service, and Responsible AI License.

Already have an account? Log in

註冊/登入

Sign up

勾選

免費帳號使用限制

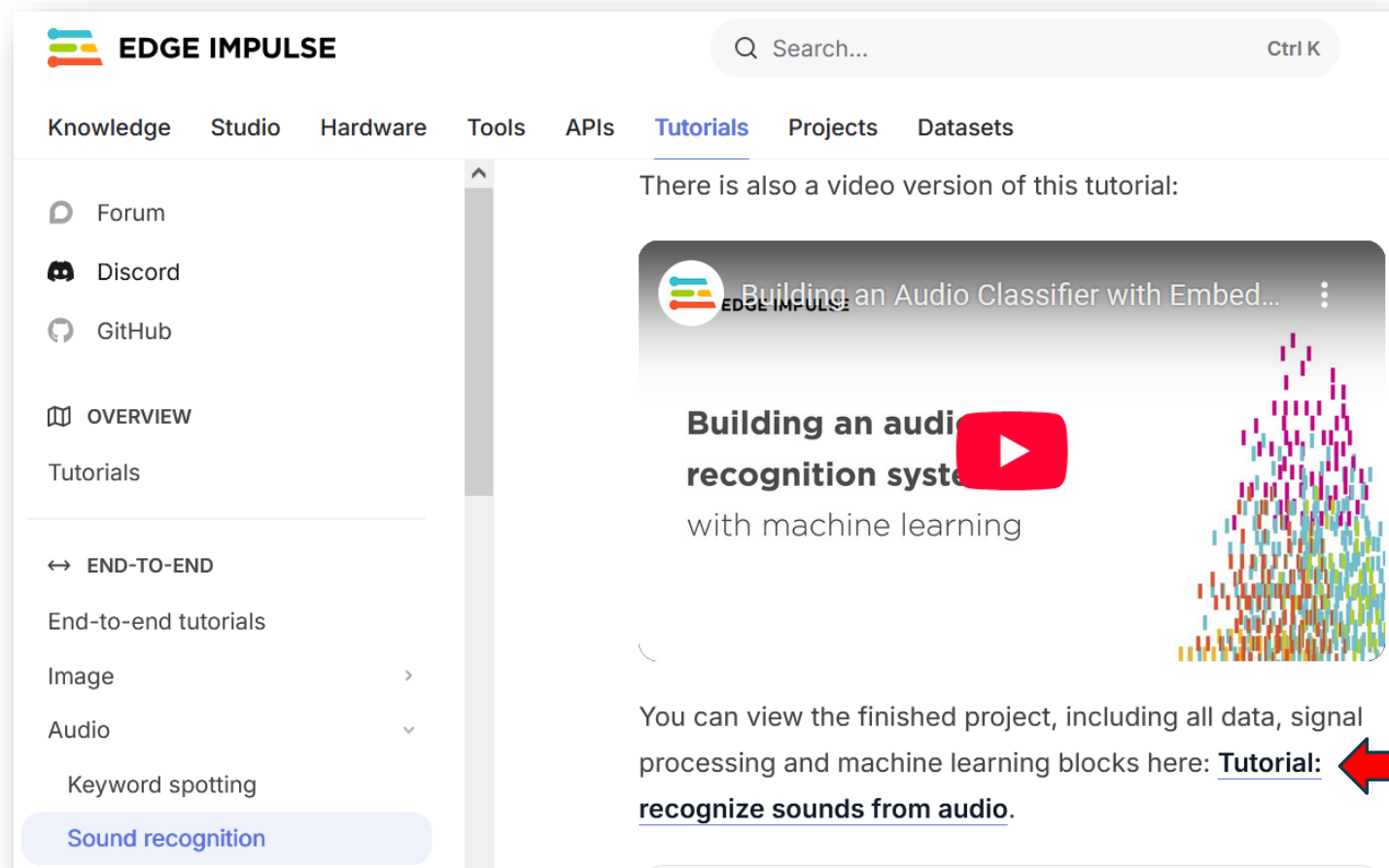
- 3組私人專案
- 公用專案不限
- 最多3個成員
- 每個專案 CPU 計算用記憶體 16GB 、計算時間60分鐘，可使用 GPU 訓練
- 開放 FOMO, FOMO-AD, GMM AD等模型，可支持 Multi-Label。
- EON Tuner, Compiler, API 呼叫有限制

<https://studio.edgeimpulse.com/login>

<https://edgeimpulse.com/pricing>

啟動公用專案範例 — 聲音辨識

Docs / Tutorials / End to End / Audio Tutors / Sound recognition

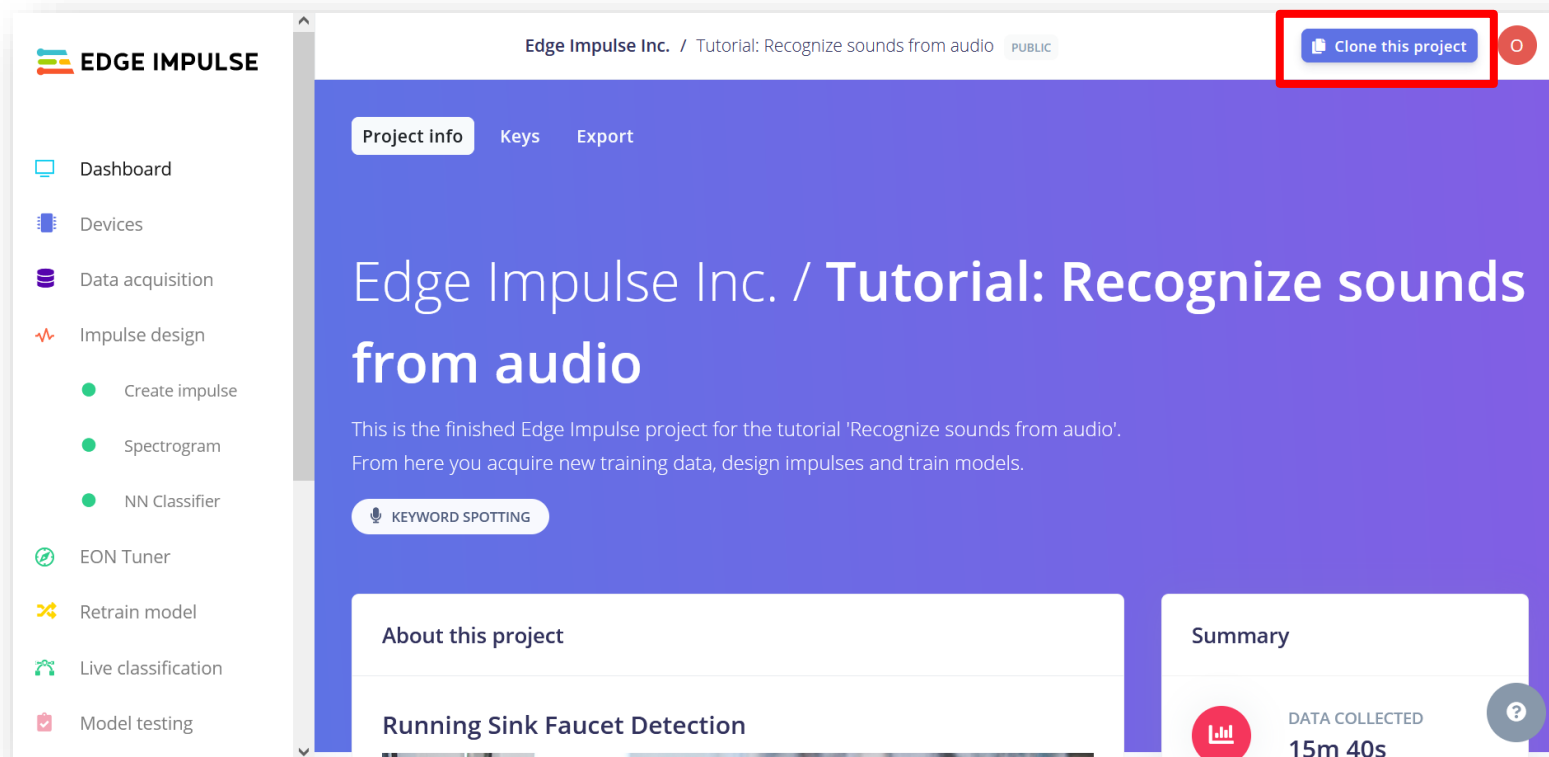


開啟範例

<https://docs.edgeimpulse.com/tutorials/end-to-end/sound-recognition>

公用專案 — 水流聲辨識

Tutorial: Recognize sounds from audio



先登入個人帳號
再複製專案到
個人帳號

水流聲辨識



<https://studio.edgeimpulse.com/public/14301/latest>

專案工作流程



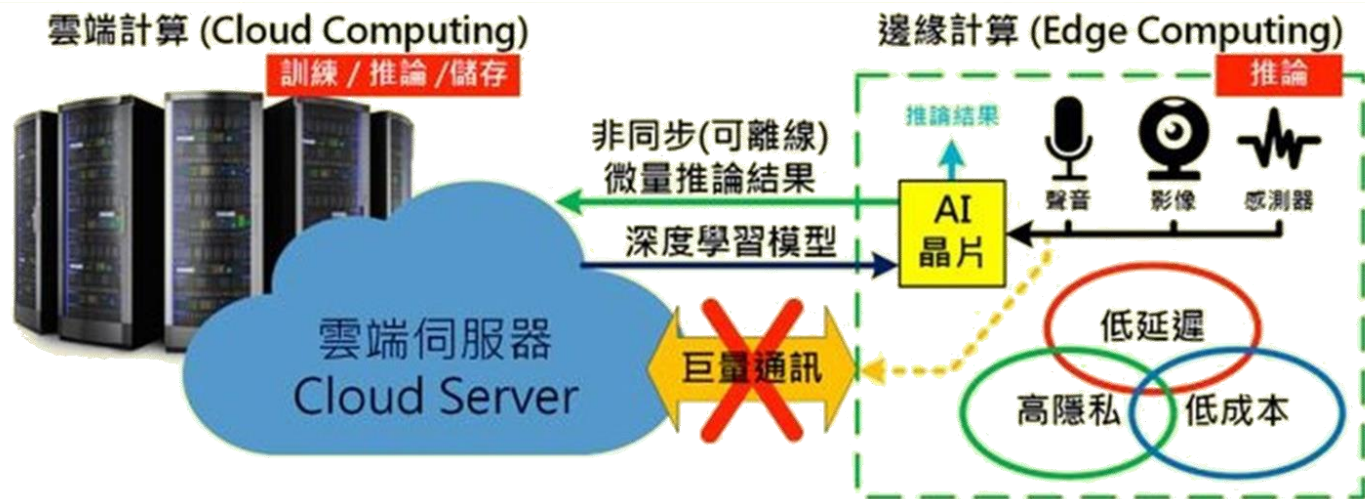
水流聲辨識

1. 準備開發板
2. 收集資料
3. 建立資料集
4. 設計及選用模型
5. 配置MFE模塊
6. 配置神經網路
7. 分類新資料
8. 模型測試
9. 模型故障排除
10. 部署到指定裝置

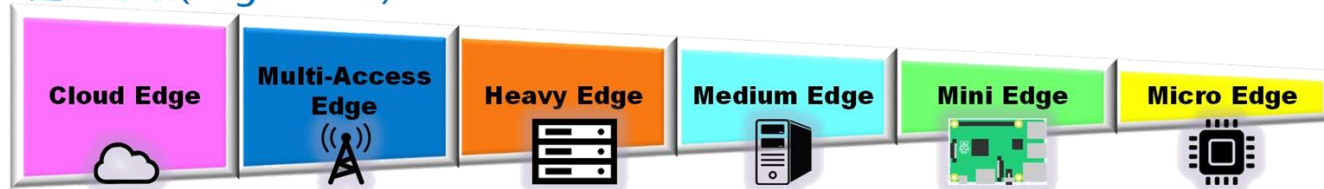
<https://docs.edgeimpulse.com/tutorials/end-to-end/sound-recognition>



Edge AI

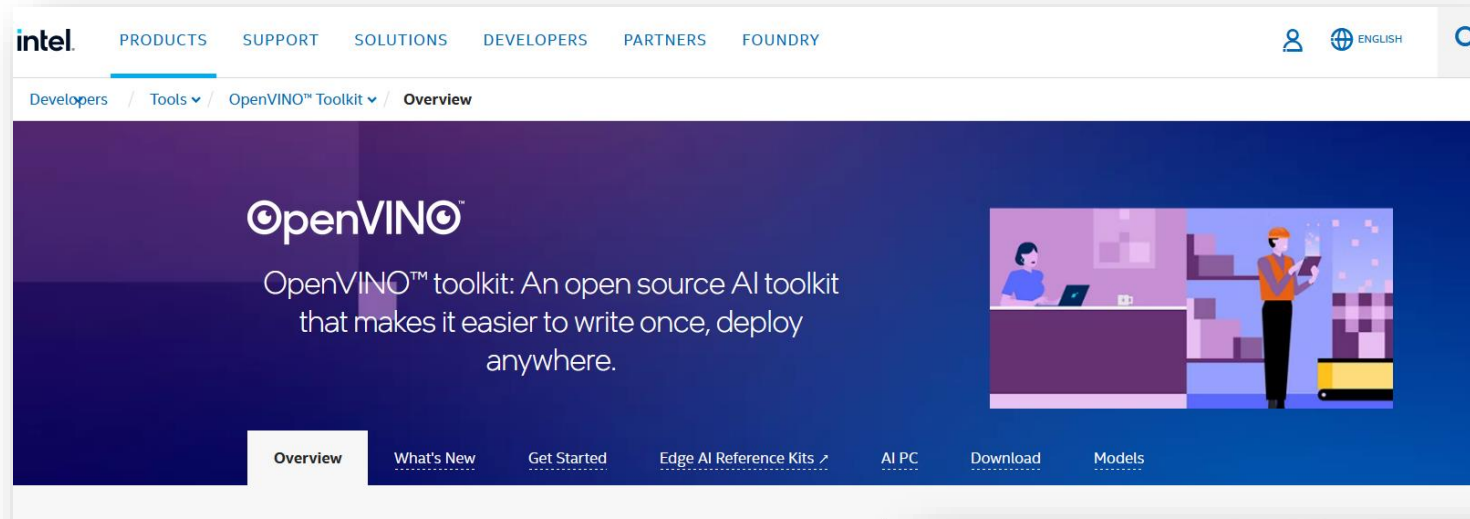


邊緣等級(Edge Level)



4.3. INTEL OPENVINO

Intel OpenVINO 官網



- Overview
- What's New
- Get Started
- Edge AI Reference Kit
- AI PC
- **Download**
- Models

Making Generative AI More Accessible for Real-World Scenarios

OpenVINO™ toolkit is an open source toolkit that accelerates AI inference with lower latency and higher throughput while maintaining accuracy, reducing model footprint, and optimizing hardware use. It streamlines AI development and integration of deep learning in domains like computer vision, large language models (LLM), and generative AI.

[What's New in 2026.0](#)

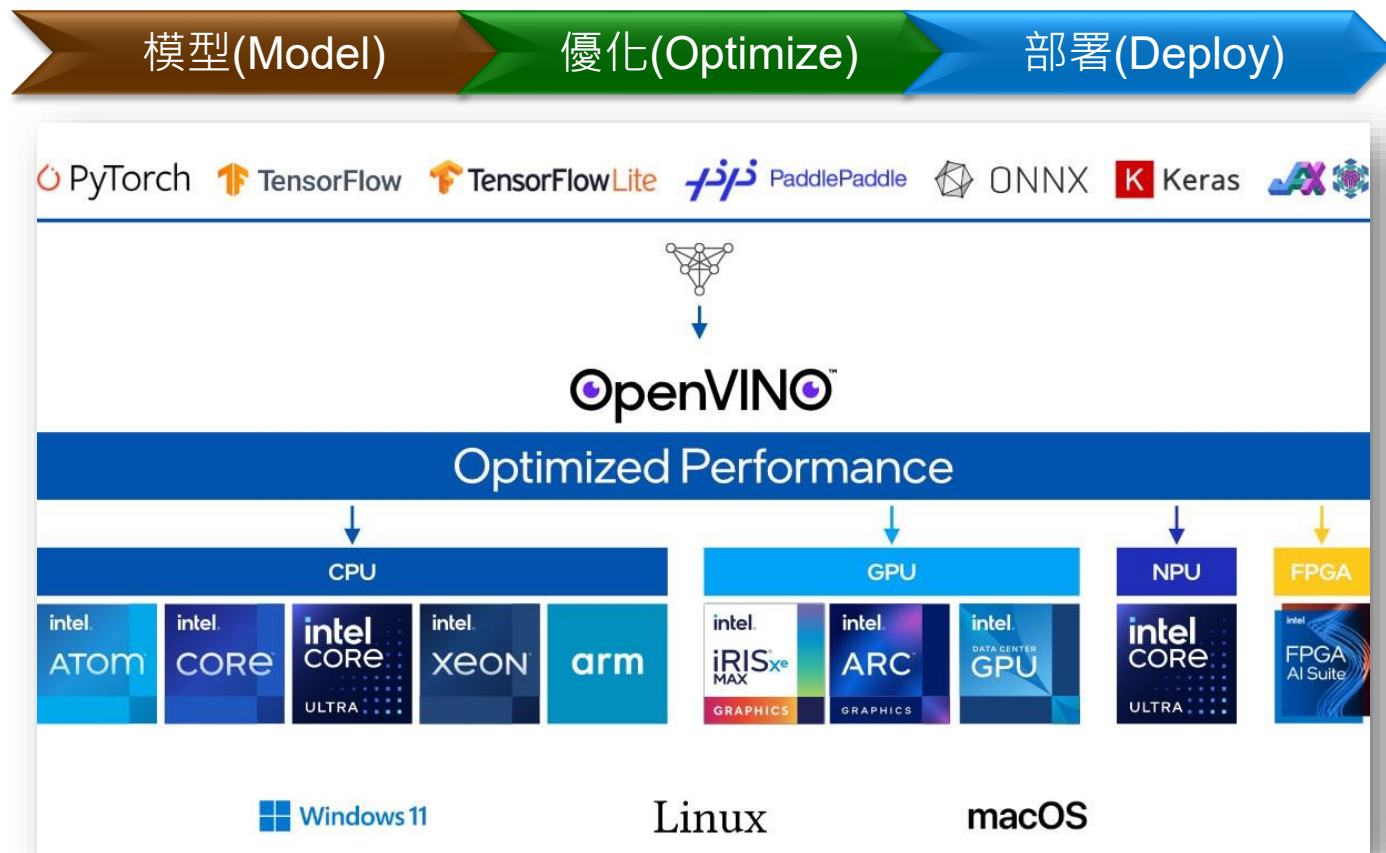
[Free Download](#)

Package	OpenVINO Base Package	OpenVINO GenAI	OpenVINO Model Server
Version	2026.0.0 Recommended	2025.4.1	Nightly Build
Operating System	Windows	macOS	Linux
Distribution	PIP Includes NPU plugin Python API only	Archives Includes NPU plugin	GitHub Source
	Conda	vcpkg Source	Gitee Source
		Conan	Docker
			npm JavaScript API only

<https://www.intel.com/content/www/us/en/developer/tools/opencvino-toolkit/overview.html>

Intel OpenVINO 說明文件

Open Visual Inference and Neural Network Optimization



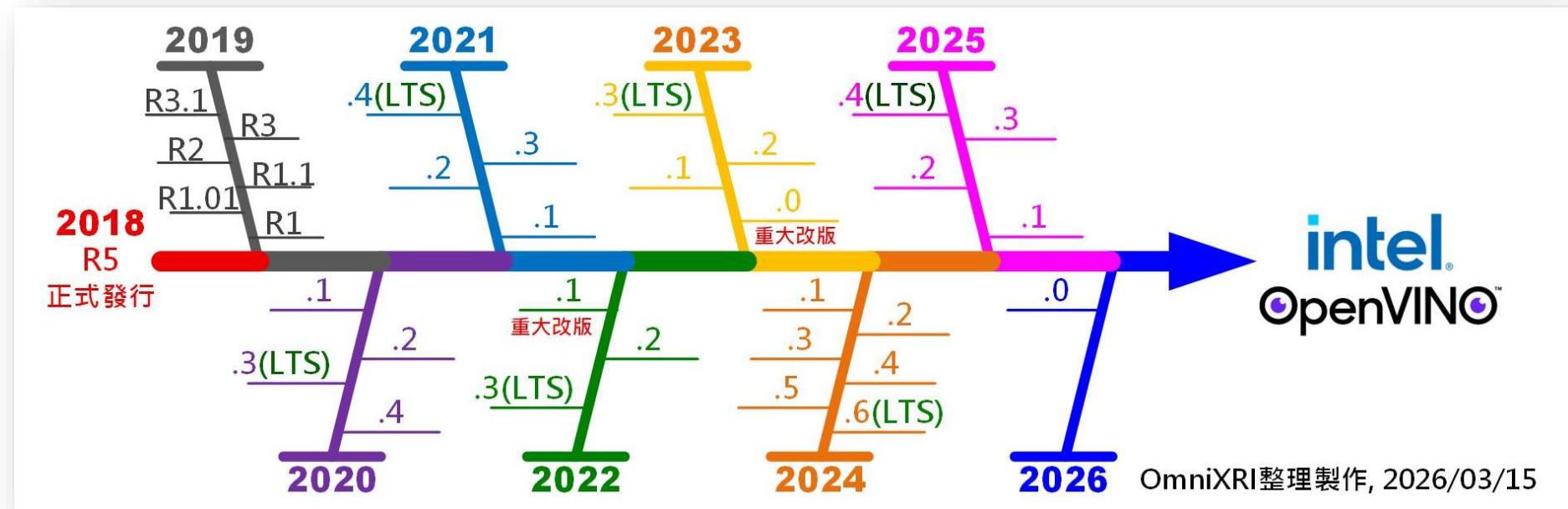
- 推論專用工具
- 跨作業系統
- 跨語言開發
- 跨AI框架
- 異質整合
- 模型優化
- 適用鑑別式及生成式AI應用

<https://docs.openvino.ai/2025/index.html>

Intel OpenVINO 演進歷史

Open Visual Inference and Neural network Optimization Toolkit

簡稱**OpenVINO**，是由Intel於2018年5月開源的一套AI推論工具包，可跨硬體支援自家的CPU, iGPU, dGPU, NPU, FPGA，至今已發行33個版本。通常一季會更新一次，一年會出一版穩定版本(Long-term support, LTS)，以確保軟體可靠性。



Intel OpenVINO Github



Open-source software toolkit for optimizing and deploying deep learning models.

[Documentation](#) • [Blog](#) • [Key Features](#) • [Tutorials](#) • [Integrations](#) • [Benchmarks](#) • [Generative AI](#)

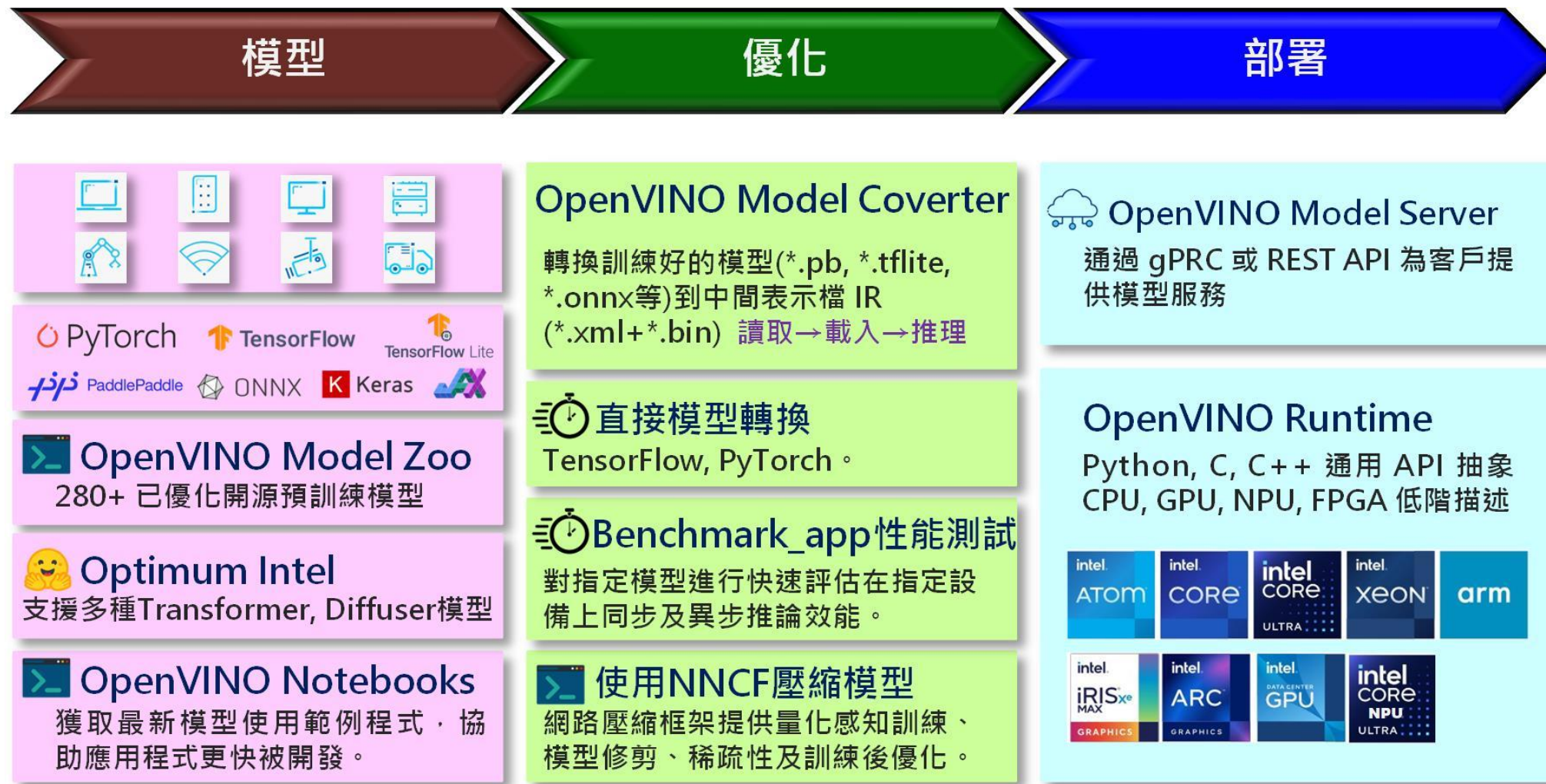
pypi package 2026.0.0 Anaconda.org 2026.0.0 homebrew v2026.0.0

downloads 23M downloads 5M total downloads 103k/year

- **Inference Optimization:** Boost deep learning performance in computer vision, automatic speech recognition, generative AI, natural language processing with large and small language models, and many other common tasks.
- **Flexible Model Support:** Use models trained with popular frameworks such as PyTorch, TensorFlow, ONNX, Keras, PaddlePaddle, and JAX/Flax. Directly integrate models built with transformers and diffusers from the Hugging Face Hub using Optimum Intel. Convert and deploy models without original frameworks.
- **Broad Platform Compatibility:** Reduce resource demands and efficiently deploy on a range of platforms from edge to cloud. OpenVINO™ supports inference on CPU (x86, ARM), GPU (Intel integrated & discrete GPU) and AI accelerators (Intel NPU).
- **Community and Ecosystem:** Join an active community contributing to the enhancement of deep learning performance across various domains.

<https://github.com/openvinotoolkit/openvino/>

Intel OpenVINO 相關工具程式



資料來源：<https://www.intel.com/content/www/us/en/developer/tools/opencvino-toolkit/overview.html>

Intel OpenVINO Model Zoo

OpenVINO™ Toolkit - Open Model Zoo repository

Note

Open Model Zoo is in maintenance mode as a source of models. Check out model tutorials in [Jupyter notebooks](#).

This repository includes optimized deep learning models and a set of demos to expedite development of high-performance deep learning inference applications. Use these free pre-trained models instead of training your own models to speed-up the development and production deployment process.

Intel is committed to the respect of human rights and avoiding complicity in human rights abuses, a policy reflected in the [Intel Global Human Rights Principles](#). Accordingly, by accessing the Intel material on this platform you agree that you will not use the material in a product or application that causes or contributes to a violation of an internationally recognized human right.

Repository Components:

- [Intel Pre-Trained Models](#)
- [Public Pre-Trained Models](#)
- [Model Downloader](#) and other automation tools
- [Demos](#) that demonstrate models usage with OpenVINO™ Toolkit
- [Accuracy Checker](#) tool for models accuracy validation

傳統 CNN, NLP 相關模型

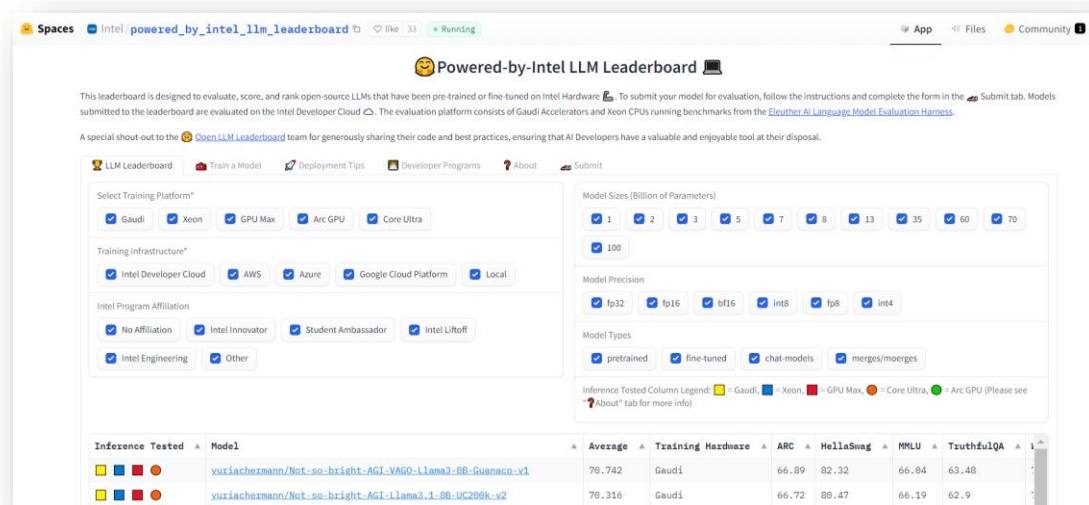
- Intel Pre-Trained Models
- Public Pre-Trained Models
- Model Downloader and other automation tools
- Demos
- Accuracy Checker

OpenVINO 2023.3(LTS)後不再更新

https://github.com/openvinotoolkit/open_model_zoo

https://docs.openvino.ai/2023.3/model_zoo.html

Hugging Face - Optimum Intel



<https://huggingface.co/Intel>

<https://huggingface.co/docs/optimum-intel/openvino/models>

- 部落格文章
- Optimum Intel
 - Transformers, Diffusers, Timm, Sentence Transformers, OpenCLIP
- GenAI Collections (40+)
 - 文字生成、文字生成影像、聲音分類、自動語音辨識、特徵提取、影像分類、影像生成文字、遮罩式語言模型、問答、文字分類、文字到文字生成、翻譯、密集預測、語音轉文字、自然對話、影像生成 ...
- 模型排行榜
- 模型庫 (200+)
- 資料集 (17)

Intel OpenVINO Notebooks

OpenVINO™ Notebooks

license Apache 2.0 treon passing docker_treon passing

A collection of ready-to-run Jupyter notebooks for learning and experimenting with the OpenVINO™ Toolkit. The notebooks provide an introduction to OpenVINO basics and teach developers how to leverage our API for optimized deep learning inference.

Checkout interactive GitHub pages application for navigation between OpenVINO™ Notebooks content:

[OpenVINO™ Notebooks at GitHub Pages](#)

OpenVINO Notebooks

Categories AI Tasks Ecosystem

Filter Ecosystem by name

OpenVINO

NNCF

Model Converter

Benchmark Tool

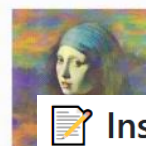
Model Server

Notebooks 21 of 149

Reset Filters

Filter notebooks by name

Sort: Recently Added



AI Trends InstantID: Zero-shot Identity-Preserving Generation using OpenVINO

Model Demos • Image-to-Image • Text-to-Image

Installation Guide

OpenVINO Notebooks require Python and Git. To get started, select the guide for your operating system or environment:

[Windows](#)

[Ubuntu](#)

[macOS](#)

[Red Hat](#)

[CentOS](#)

[Azure ML](#)

[Docker](#)

[Amazon SageMaker](#)

- 豐富實作案例
- 電腦視覺
- 自然語言處理
- 大語言模型
- 語音辨識
- 多模態模型

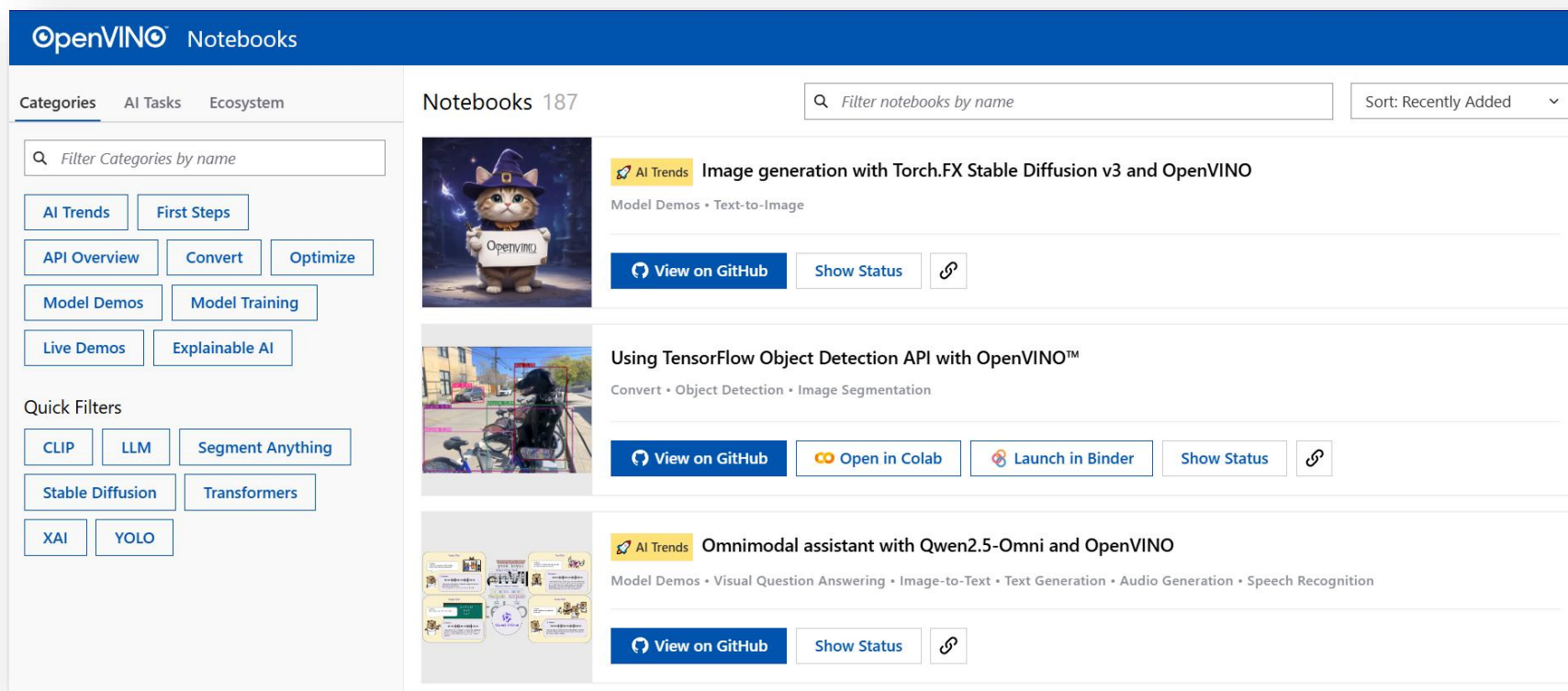
https://github.com/openvinotoolkit/openvino_notebooks

Intel OpenVINO Notebooks Pages

基於 Jupyter Notebook 型式範例

 launch binder and  Open in Colab

雲端直接
執行範例



- 豐富實作案例
- 電腦視覺
- 自然語言處理
- 大語言模型
- 語音辨識
- 多模態模型

https://openvinotoolkit.github.io/openvino_notebooks/

Intel OpenVINO 鑑別式 AI 應用

Computer Vision

Image Classification

Image Segmentation

Image Inpainting

Image-to-Image

Object Detection

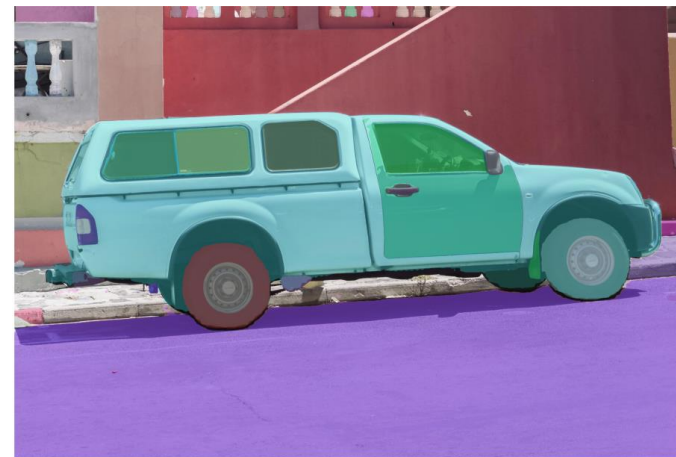
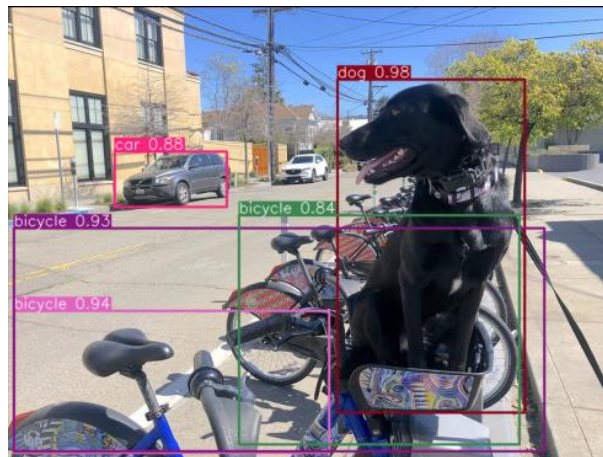
Depth Estimation

Style Transfer

Pose Estimation

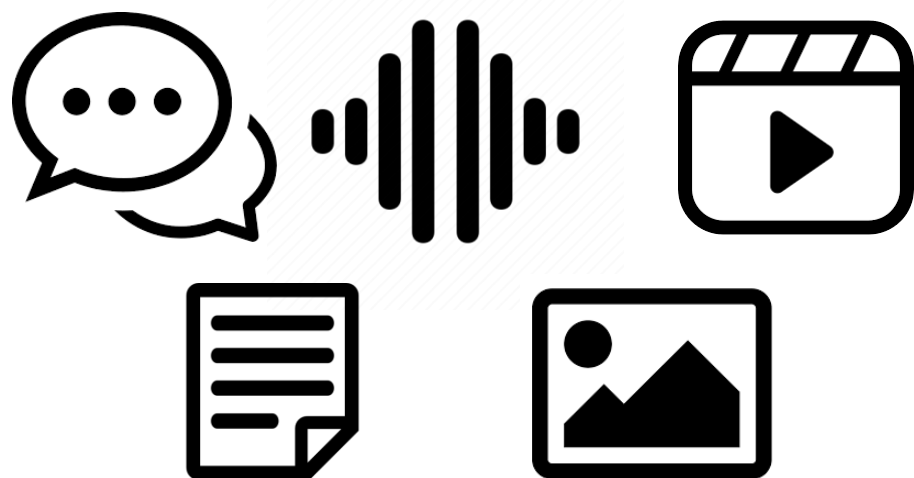
Zero-Shot Image Classification

Text Detection

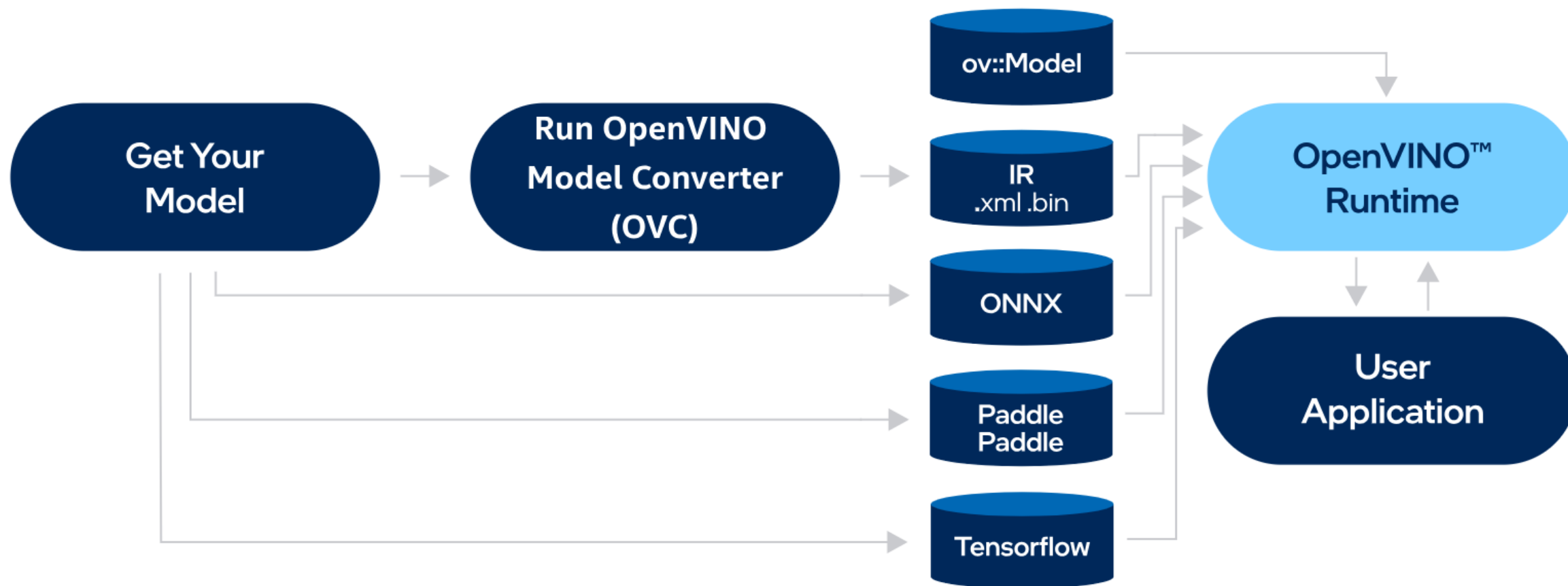


Intel OpenVINO 生成式 AI 應用

- 文字生成文字
- 文字生成影像/影片
- 文字生成語音
- 語音生成文字
- 影像生成影片



Intel OpenVINO Model Converter



<https://docs.openvino.ai/2026/openvino-workflow/model-preparation.html>

Intel OpenVINO 直接支援AI框架



PyTorch

```
import openvino.torch
#compile PyTorch model as usual with PyTorch
compiled_model = torch.compile(model, backend="openvino", options =
{ "device" : "CPU" })
```



ONNX Runtime

```
onnx_model = onnx.load("model.onnx")
onnx.save_model(onnx_model, 'saved_model.onnx')
sess.set_providers(['OpenVINOExecutionProvider'])
```



Hugging Face

```
from optimum.intel import OVModelForCausalLM
#define model_id, use transformers tokenizer & pipeline
model = OVModelForCausalLM.from_pretrained(model_id)
pipe = pipeline("text-generation", model=model, tokenizer=tokenizer)
```



LangChain

```
ov_llm=HuggingFacePipeline.from_model_id(...backend="openvino",
      model_kwargs={"device":"CPU","ov_config": ov_config})
ov_chain = prompt | ov_llm
print(ov_chain.invoke({"question":"what is neurobiology?"}))
```

Intel OpenVINO Benchmark_App

評估模型推理計算性能

OpenVINO 提供 Python, C++ 版本工具

benchmark_app -m model.xml -d GPU.1 -api async -hint latency

-m 模型名稱 (IR檔所在路徑)

-d 推論硬體 CPU, GPU (有內、外顯時以.0/.1區分) , **AUTO**, NPU

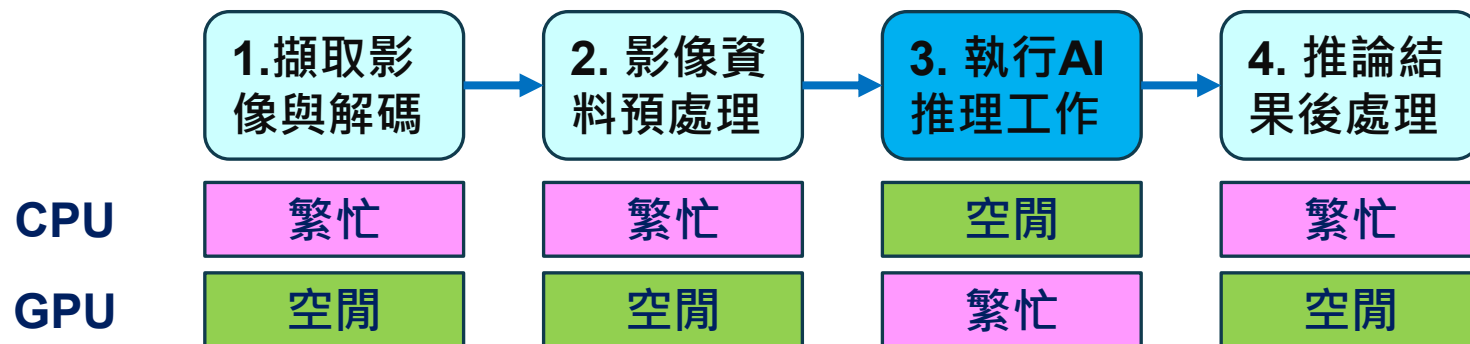
-api 指定推理模式，sync (同步) 、async (異步)

-hint 指定推理計算模式，throughput (吞吐量優先) 、cumulative_throughput (累積吞吐量優先) 、latency (延遲優先) 、none (無)

<https://docs.openvino.ai/2026/get-started/learn-openvino/openvino-samples/benchmark-tool.html>

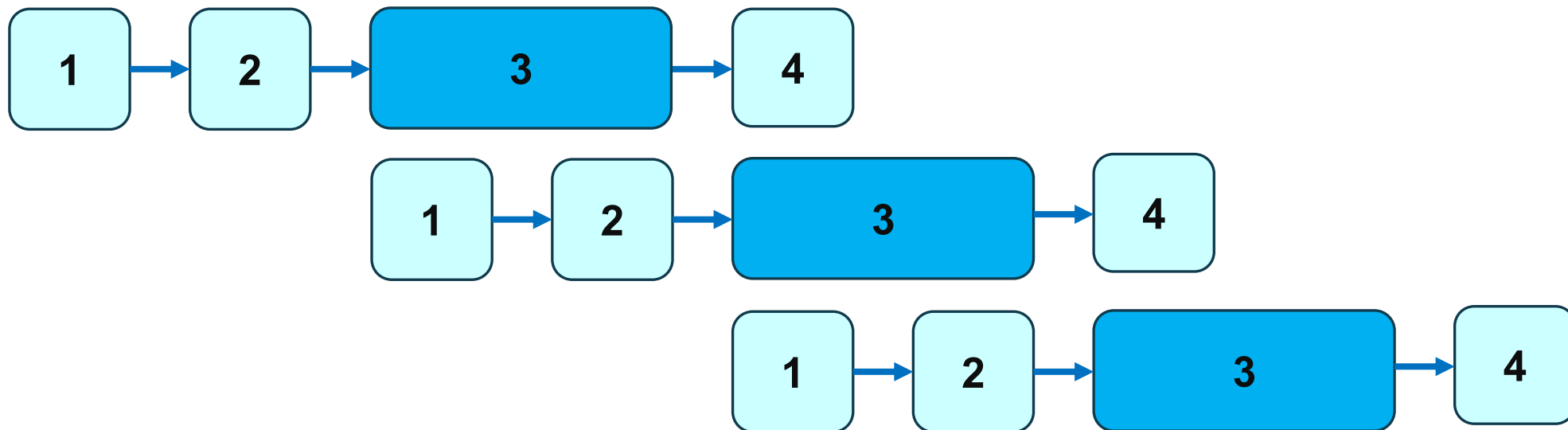
推論同步及異步處理

同步推論



補充：可使用
Preprocessing API
調用 GPU進行 1, 2, 4

異步推論



Intel OpenVINO - NNCF

Post-Training Compression Algorithms

Compression algorithm	OpenVINO	PyTorch	TensorFlow	ONNX
Post-Training Quantization	Supported	Supported	Supported	Supported
Weights Compression	Supported	Supported	Not supported	Not supported
Activation Sparsity	Not supported	Experimental	Not supported	Not supported

可支援量化數值格式
FP32, FP16, BF16, INT8, INT4

模型壓縮
推論加速
精度不變

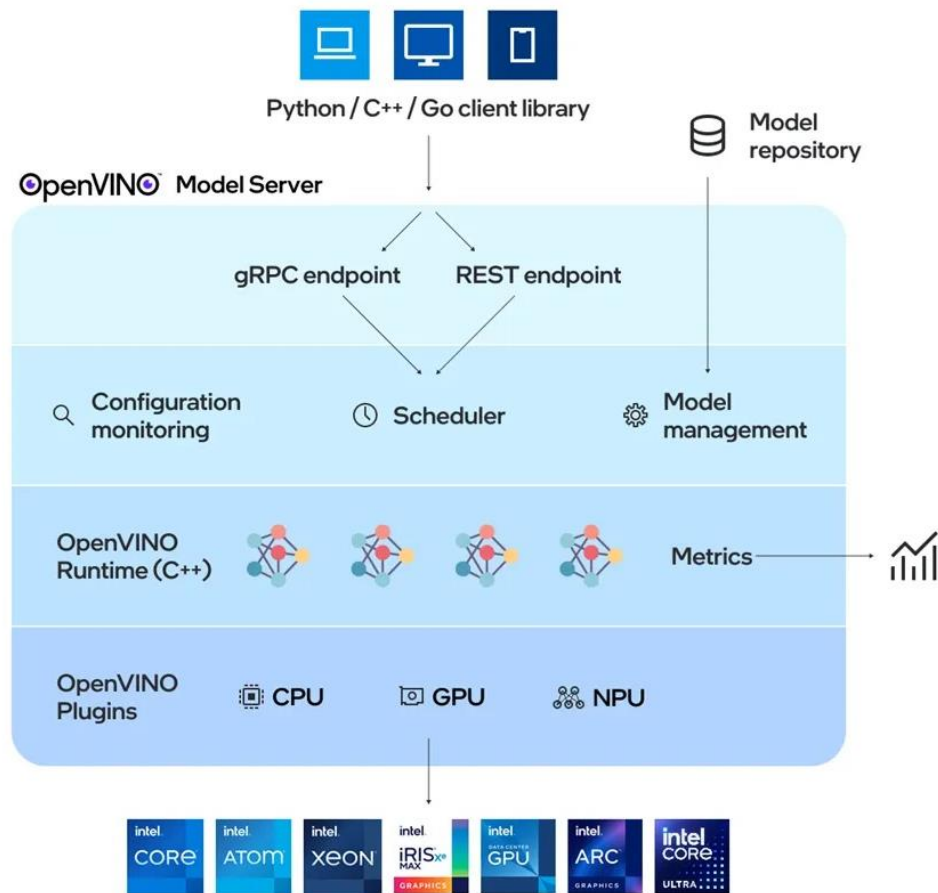
Training-Time Compression Algorithms

Compression algorithm	PyTorch	TensorFlow
Quantization Aware Training	Supported	Supported
Mixed-Precision Quantization	Supported	Not supported
Sparsity	Supported	Supported
Filter pruning	Supported	Supported
Movement pruning	Experimental	Not supported



<https://github.com/openvinotoolkit/nncf>

Intel OpenVINO Model Server



1. 極致的硬體加速效能

- 異構運算支援
- 針對 GenAI 優化

2. 標準化與相容性

- 相容主流框架
- 多模型格式支援

3. 雲端原生與擴充性

- Kubernetes 整合
- 容器化部署

4. 進階管理功能

- 動態輸入支援
- 模型版本管理
- 監控與指標

https://docs.openvino.ai/2026/model-server/ovms_what_is_openvino_model_server.html

Intel OpenVINO Runtime

Install OpenVINO™ Runtime on Linux

If you want to install OpenVINO™ Runtime on Linux, you have the following options:

- [Install OpenVINO using an Archive File](#)
- [Install OpenVINO using PyPI](#)
- [Install OpenVINO using APT](#)
- [Install OpenVINO using YUM](#)
- [Install OpenVINO using ZYPPE](#)
- [Install OpenVINO using Conda Forge](#)
- [Install OpenVINO using vcpkg](#)
- [Install OpenVINO using Homebrew](#)
- [Install OpenVINO using Docker](#)
- [Install OpenVINO using Conan Package Manager](#)
- [Install OpenVINO using npm](#)

Install OpenVINO™ Runtime on Windows

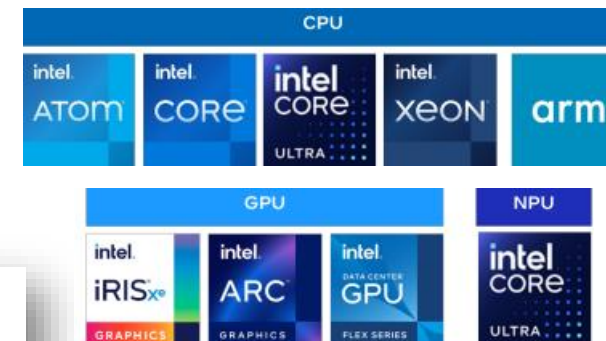
If you want to install OpenVINO™ Runtime on Windows, you have the following options:

- [Install OpenVINO using an Archive File](#)
- [Install OpenVINO using PyPI](#)
- [Install OpenVINO using Conda Forge](#)
- [Install OpenVINO using vcpkg](#)
- [Install OpenVINO using Docker](#)
- [Install OpenVINO using Conan Package Manager](#)
- [Install OpenVINO using npm](#)

Install OpenVINO™ Runtime for macOS

If you want to install OpenVINO™ Runtime on macOS, you have the following options:

- [Install OpenVINO using an Archive File](#)
- [Install OpenVINO using PyPI](#)
- [Install OpenVINO using Conda Forge](#)
- [Install OpenVINO using Homebrew](#)
- [Install OpenVINO using vcpkg](#)
- [Install OpenVINO using Conan Package Manager](#)
- [Install OpenVINO using npm](#)



<https://docs.openvino.ai/2026/get-started/install-openvino/install-openvino-linux.html>
<https://docs.openvino.ai/2026/get-started/install-openvino/install-openvino-windows.html>
<https://docs.openvino.ai/2026/get-started/install-openvino/install-openvino-macos.html>

Intel OpenVINO – Gen AI API

Install OpenVINO™ 2024.2

2024.2 後支援

Package	OpenVINO Base Package Full inference with basic GenAI	OpenVINO with GenAI Full inference + complete GenAI package
Version	2024.2 Recommended	
Operating System	Windows	macOS
Distribution	OpenVINO Archives Includes NPU plugin	PIP Includes NPU plugin Python API only
Install	# Use the following Link: Download Archives with GenAI	

	OpenVINO GenAI API	Optimum-Intel
Additional Dependencies	Very lightweight	Many dependencies, including PyTorch, etc.
Inference code	Very light coding	Hugging Face based
APIs	Python, C++	Python
Models support	LLMs (additional generative models coming soon)	Broad set of models
Sampling methods	Greedy, Multinomial and Beam search	Full (Speculative Sampling and Contrastive search)
Parallel inference optimization	Yes	No
Tokenization	via OpenVINO tokenizer models	via Hugging Face tokenizer

`pip install optimum[openvino]`

```

from optimum.intel import OVModelForCausalLM
from transformers import AutoConfig, AutoTokenizer

model_dir = r"llama-3-8b-instruct\INT4_compressed_weights"

ov_config = {'PERFORMANCE_HINT': 'LATENCY', 'NUM_STREAMS': '1', "CACHE_DIR": ""}
chat_model = OVModelForCausalLM.from_pretrained(model_dir, device="AUTO",
                                                config=AutoConfig.from_pretrained(model_dir),
                                                ov_config=ov_config)

chat_tokenizer = AutoTokenizer.from_pretrained(model_dir)

prompt="The Sun is yellow because"
inputs = chat_tokenizer(prompt, return_tensors="pt").to(chat_model.device)
input_length = inputs.input_ids.shape[1]
outputs = chat_model.generate(**inputs, max_new_tokens=256,
                             do_sample=True, temperature=0.6, top_p=0.9, top_k=50)
tokens = outputs[0, input_length:]
print(chat_tokenizer.decode(tokens, skip_special_tokens=True))

```

OpenVINO™ Integration with Optimum

`pip install openvino-genai`

```

import openvino_genai as ov_genai
pipe = ov_genai.LLMPipeline(model_path, "AUTO")
print(pipe.generate("The Sun is yellow because"))

```

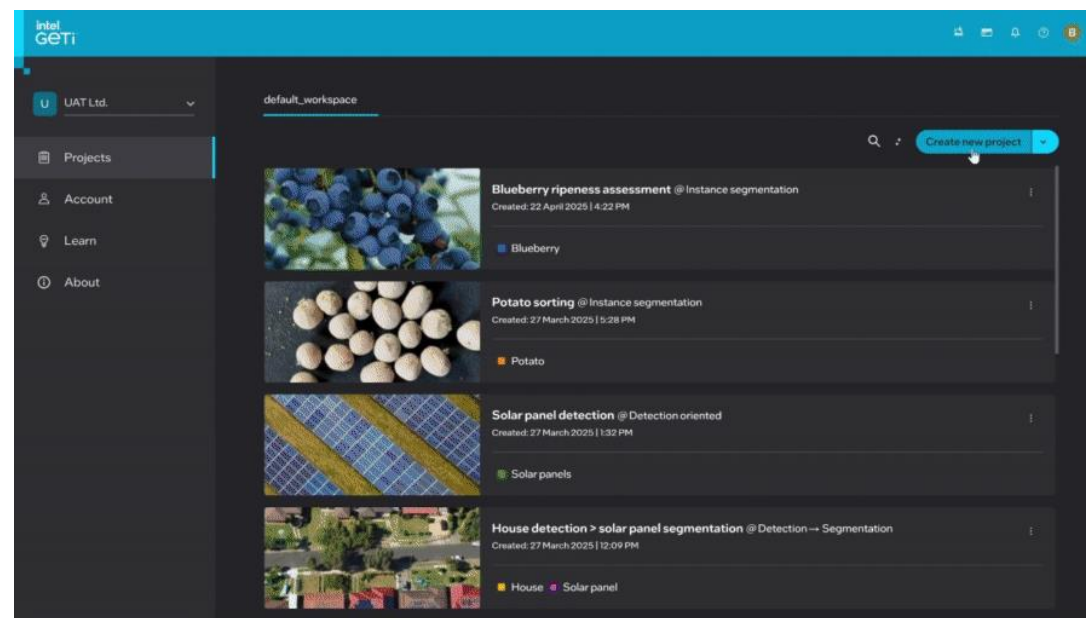
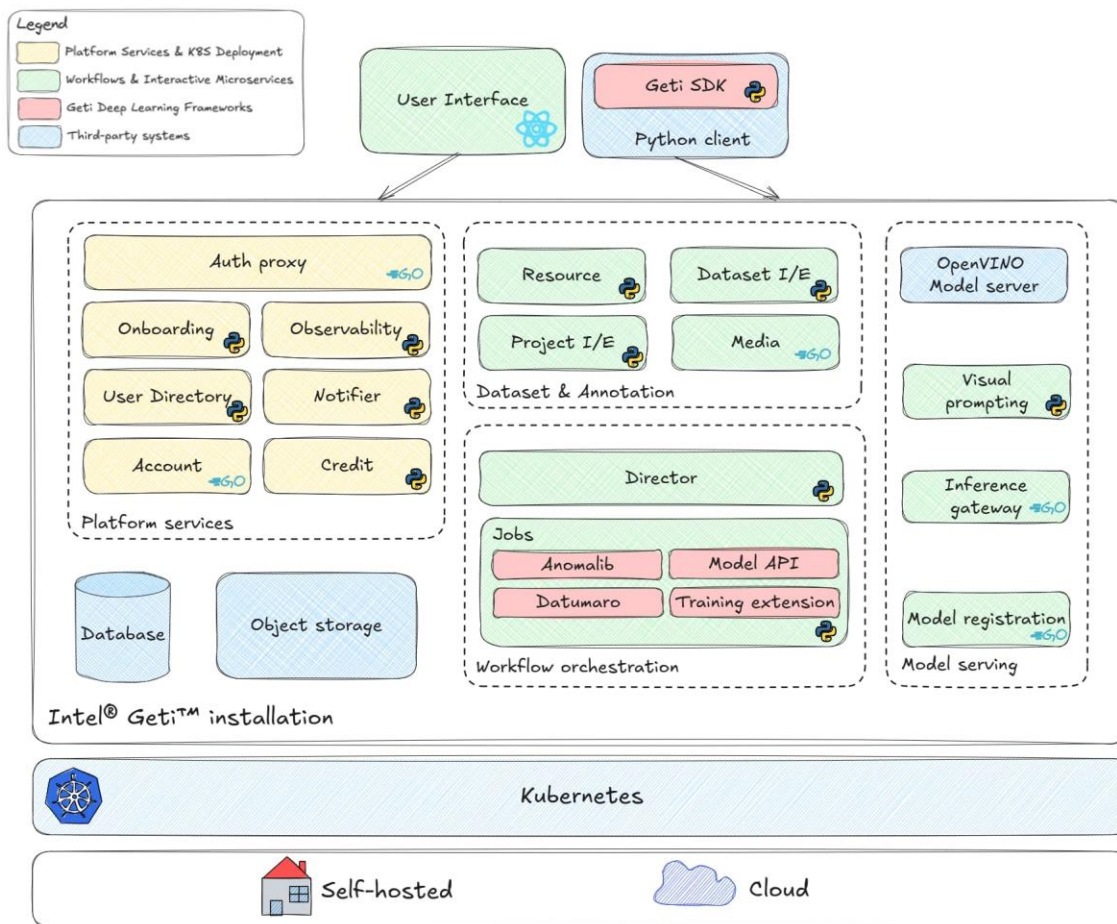
OpenVINO™ Gen AI API

<https://docs.openvino.ai/2026/get-started/install-openvino/install-openvino-genai.html>

Intel 快速建構部署模型工具 — Geti

(免費開源工具、可雲、可端)

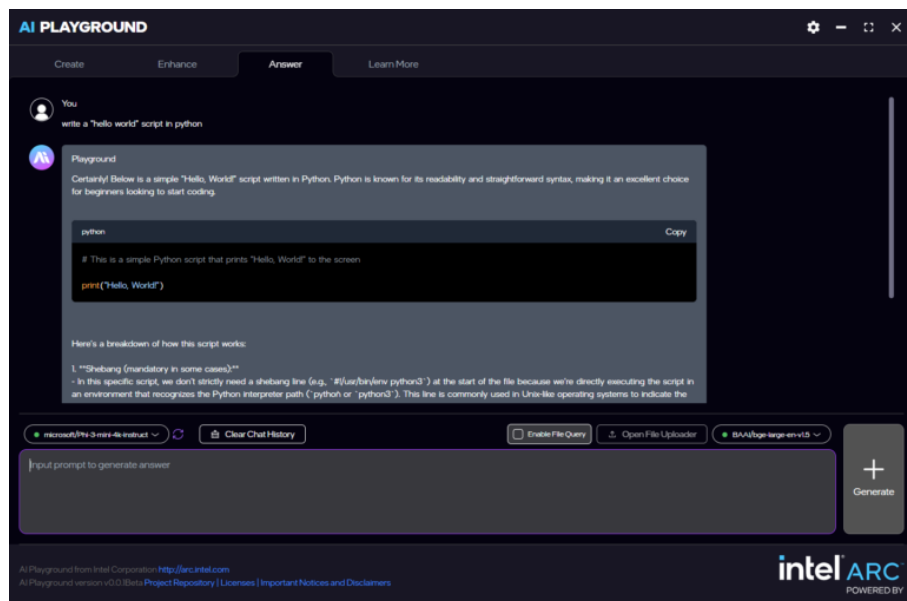
透過圖形化介面可以快速添加資料及標註、模型快速訓練、優化及部署，並可輕鬆整合 OpenVINO。



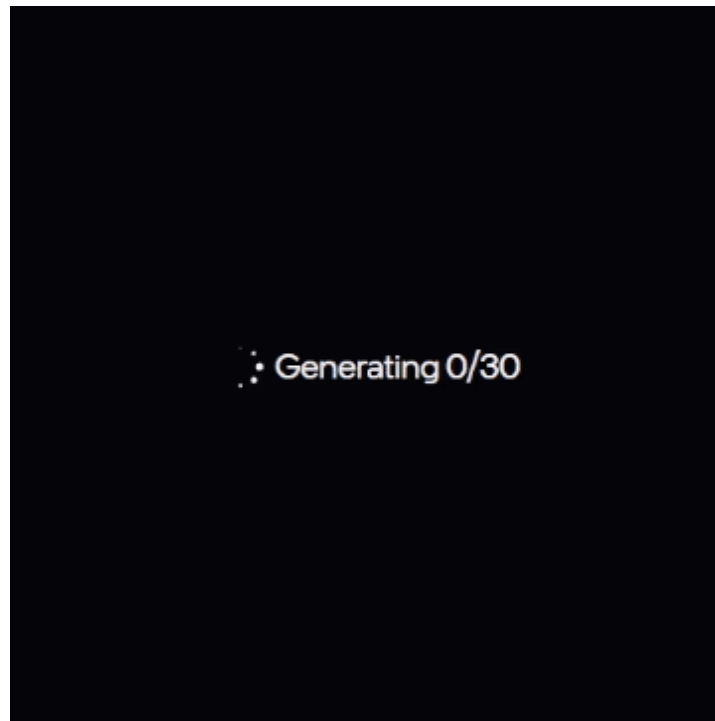
資料來源：<https://github.com/open-edge-platform/geti>

Intel AI Playground 生成式 AI 工具

聊天機器人



影像生成



影像增強



適用硬體：Arc dGPU 8GB+, Core Ultra-H PC, Core Ultra 200V Series (註一)

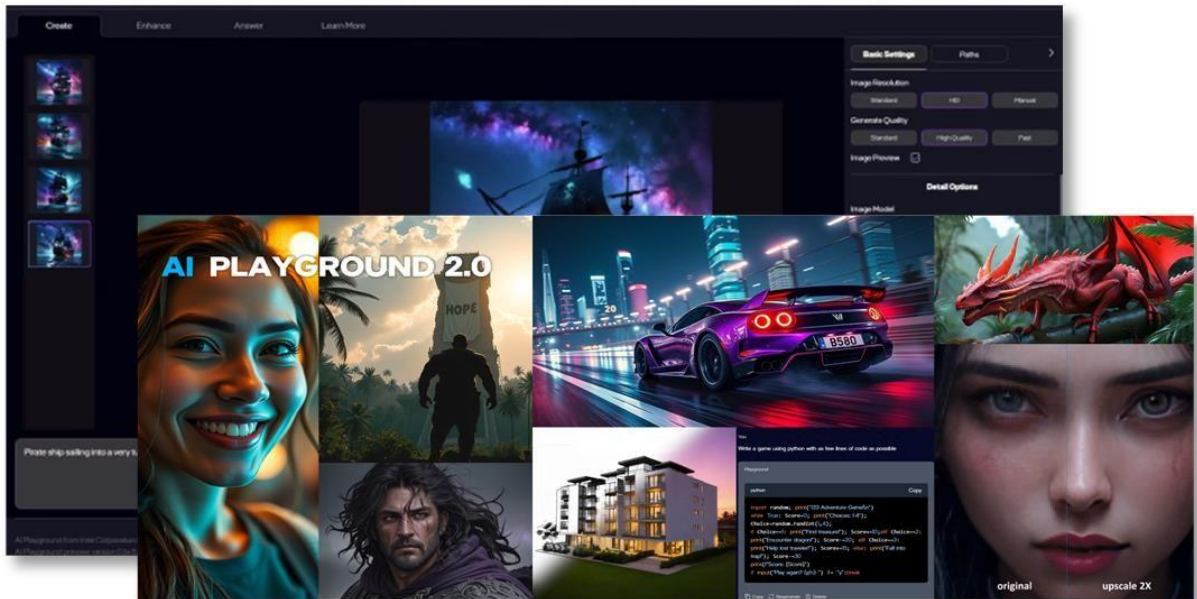
直接下載模型 免寫程式 即視即用

資料來源：<https://game.intel.com/tw/stories/introducing-ai-playground/>

註一：Core Ultra 200V Coming Soon

Intel AI Playground 展示

Create Enhance Answer



AI PLAYGROUND 2.0

OpenVINO™ GenAI

Lunar Lake
Core Ultra 200
Arc GPU + NPU

有了 **Intel AI Playground**
不寫程式也能輕鬆玩生成式AI

<https://omnixri.blogspot.com/2025/09/intel-ai-playground-ai.html>

Intel OpenVINO Notebooks 安裝(1/2)

1. 安裝Python / Git

(Windows 版本安裝)

至[Python官網](#)下載並安裝對應版本原則上支援3.9 ~ 3.11版 Python 64bit

至[Git官網](#)下載並安裝最新版本版本

2. 安裝 GPU / NPU 最新驅動程式 (非AI PC可略)

3. 安裝C++ Redistributable(必要), FFMPEG (選用)

下載[Microsoft Visual C++ Redistributable](#)並安裝。

步驟1~3為執行
Notebooks 前必
要動作，若已安
裝過則可略過。

4. 開啟命令列模式，建立 Python 虛擬環境

`python -m venv openvino_env`

5. 啟重虛擬環境

`openvino_env\Scripts\activate`

(可支援作業系統)

[Windows](#)

[Ubuntu](#)

[macOS](#)

[Red Hat](#)

[CentOS](#)

[Azure ML](#)

[Docker](#)

[Amazon SageMaker](#)

https://github.com/openvinotoolkit/openvino_notebooks/wiki/Windows

Intel OpenVINO Notebooks 安裝(2/2)

6. 複製程式庫並進入工作路徑

```
git clone --depth=1 https://github.com/openvinotoolkit/openvino_notebooks.git  
cd openvino_notebooks
```

7. 安裝相關套件

```
python -m pip install --upgrade pip wheel setuptools  
pip install -r requirements.txt
```

若沒安裝過 OpenVINO 會自動安裝。若已有安裝過 OpenVINO 則會自動移除後再自動重新安裝。

8. 執行範例

若欲執行所有範例則在命令列執行「**jupyter lab notebooks**」，進入後再選擇欲運行的範例。

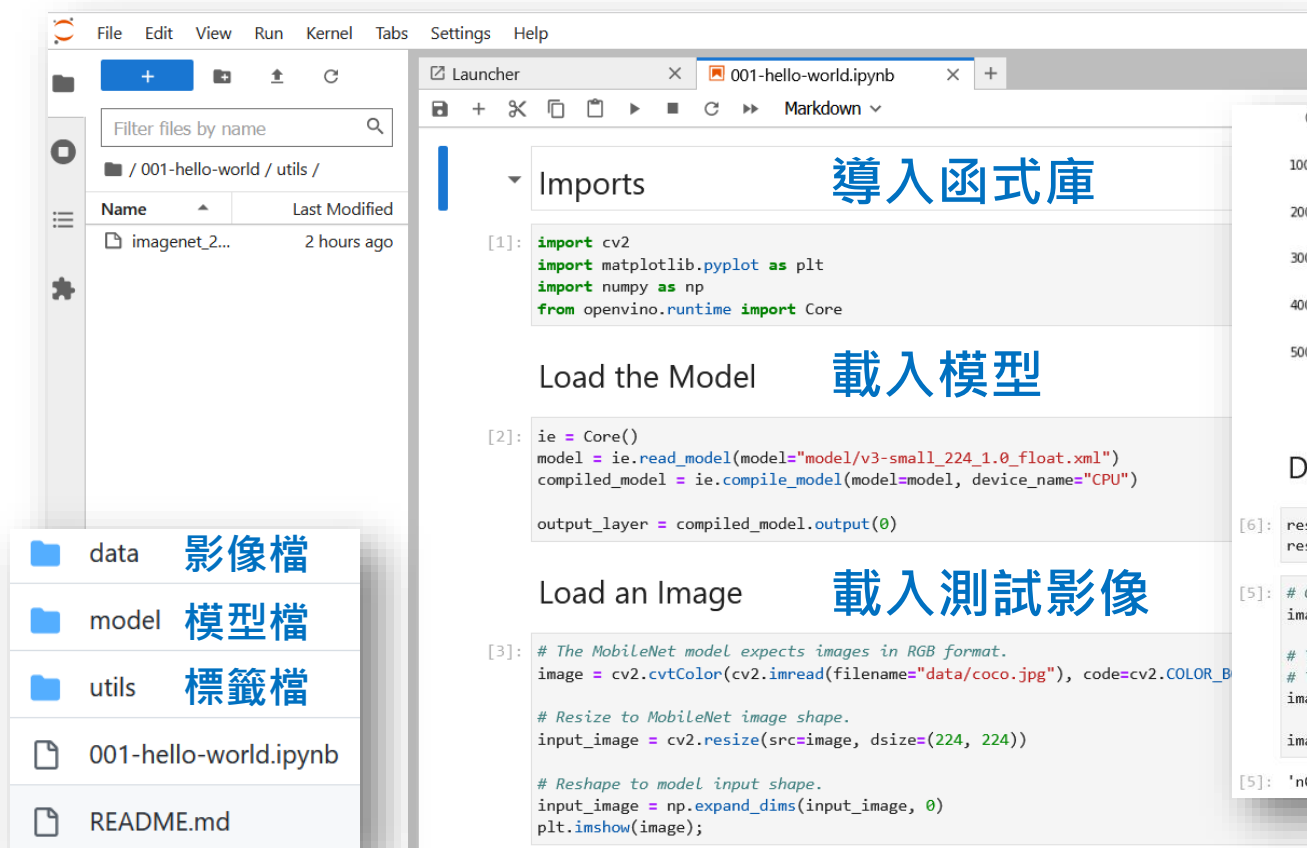
若想運行特定範例則指定對應程式路徑及名稱(*.ipynb)即可。

```
jupyter notebook notebooks/hello-world/hello-world.ipynb
```

https://github.com/openvinotoolkit/openvino_notebooks/wiki/Windows

Notebooks範例 - hello-world

Jupyter Notebook 操作介面



平毛尋回犬
(Flat-coated
Retriever)



進行推論

Do Inference

```
[6]: result_infer = compiled_model([input_image])[output_layer]
result_index = np.argmax(result_infer)

[5]: # Convert the inference result to a class name.
imagenet_classes = open("utils/imagenet_2012.txt").read().splitlines()

# The model description states that for this model, class 0 is a background.
# Therefore, a background must be added at the beginning of imagenet_classes.
imagenet_classes = ['background'] + imagenet_classes

imagenet_classes[result_index]

[5]: 'n02099267 flat-coated retriever'
```

在命令視窗按**Ctrl+C**結
束Jupyter Notebook

Intel OpenVINO Notebooks Hello範例



Hello Object Detection
First Steps • Object Detection

物件偵測

[View on GitHub](#) [Open in Colab](#) [Launch in Binder](#) [Show Status](#) [Link](#)



Hello Image Segmentation
First Steps • Image Segmentation

影像分割

[View on GitHub](#) [Open in Colab](#) [Launch in Binder](#) [Show Status](#) [Link](#)



Hello Image Classification
First Steps • Image Classification

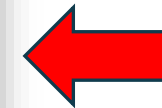
影像分類

[View on GitHub](#) [Open in Colab](#) [Launch in Binder](#) [Show Status](#) [Link](#)

Google Colab 範例
搜尋欄輸入 hello

[Open in Colab](#)

可直接於Colab執行



https://openvinotoolkit.github.io/openvino_notebooks/?search=hello

Google Colab 雲端執行 OpenVINO

hello-world.ipynb 工作流程

- 導入必要函式庫 (import)
- 下載模型和測試影像
- 選擇推論裝置 (Colab 只能選CPU)
- 載入模型
- 載入影像
- 執行推論

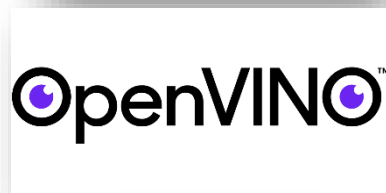
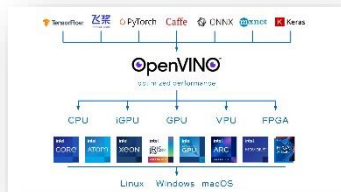
執行結果



```
imagenet_classes[result_index]  
  
'n02099267 flat-coated retriever'
```

https://colab.research.google.com/github/openvinotoolkit/openvino_notebooks/blob/latest/notebooks/hello-world/hello-world.ipynb

Intel OpenVINO 學習資源



- Intel OpenVINO Document

<https://docs.openvino.ai/>

- Intel OpenVINO Toolkit (Youtube)

<https://www.youtube.com/watch?v=kY9nZbX1DWM&list=PLg-UKERBljNxdIQir1wrirZJ50yTp4eHv>

- Intel OpenVINO Toolkit (Medium)

<https://medium.com/@openvino>

- OmniXRI，許哲豪，OpenVINO系列發文

https://hackmd.io/@OmniXRI-Jack/series_articles#OpenVINO%E7%B3%BB%E5%88%97

- MakerPRO，OpenVINO系列發文

<https://makerpro.cc/category/suppliers/openvino-column/>

參考文獻

- 許哲豪，臺灣科技大學資訊工程系 人工智慧與邊緣運算實務 (CS5149701)
<https://omnixri.blogspot.com/p/ntust-edge-ai.html>
- Arm,通用微控制器軟體介面標準 (CMSIS)
<https://www.arm.com/zh-tw/technologies/cmsis>
- 許哲豪，OmniXRI TinyML 小學堂 (2025) 【第 5 講】通用微控制器軟體介面標準(CMSIS)
<https://youtu.be/JFR5o4dFxrK>
- 許哲豪，TinyML 核心函式庫 Arm CMSIS 6 DSP & NN 更新比較
<https://omnixri.blogspot.com/2024/02/tinyml-arm-cmsis-6-dsp-nn.html>
- Edge Impulse Documentation
<https://docs.edgeimpulse.com/>
- Intel OpenVINO Document
<https://docs.openvino.ai/>
- Intel OpenVINO Notebooks (Github)
https://github.com/openvinotoolkit/openvino_notebooks